

Mechanistic Insights into Aldehyde Production from Electrochemical CO₂ Reduction on CuAg Alloy via *Operando* X-Ray Measurements

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ABSTRACT. CO₂ electrolysis converts the greenhouse gas CO₂ into valuable fuels and chemicals, such as carbon monoxide, ethylene, ethanol, etc. Currently, Cu is the only known monometallic catalyst capable of producing multicarbon products from electrochemical CO₂ reduction reaction (eCO₂RR), while the poor selectivity limits its further use. It has been found that introducing Ag atoms into the Cu lattice can modulate product preference. However, the synergistic effects between Cu and Ag, and thus the catalytic performance, are strongly influenced by catalyst morphology, electrolyzer configuration, reaction conditions, etc. *Operando* measurements can provide explicit information on the catalyst dynamic variation

during the reaction, but their operation and analysis are challenging. Herein, we prepared CuAg multiphase alloy catalysts by magnetron sputtering, which allowed for investigating the intrinsic interaction between Cu and Ag. eCO₂RR performance exhibited an improved selectivity towards carbonyls at the expense of hydrogen and hydrocarbons. The partially alloyed Cu and Ag phases were confirmed by *operando* X-ray diffraction. By means of combining *operando* X-ray measurements and density functional theory (DFT) calculations, the preferred carbonyl production is attributed to the reduced electron density and compressive strain of Cu due to Ag incorporation, which leads to a deeper d-band center and therefore weakened intermediate adsorption and oxophilicity. This work provides evidence of the intrinsic structural and electronic interaction between Cu and Ag during eCO₂RR. The obtained information will facilitate the design of bi-/multi-phase metallic or alloy electrocatalysts.

INTRODUCTION

Climate change driven by the continuously increasing CO₂ concentration in the atmosphere is one of the greatest threats facing humans. The Intergovernmental Panel on Climate Change (IPCC) released a special report (SR15) in 2018, proposing that in order to limit global warming to the safe limit, which is 1.5 °C above preindustrial levels, the global net CO₂ emission needs to reach "net zero" by 2050¹. Electrochemical CO₂ reduction (eCO₂RR) converts CO₂ into valuable fuels and chemicals, and thus helps with closing the anthropogenic carbon cycle. Among all the reported eCO₂RR catalysts, Cu is the only monometallic catalyst that produces both hydrocarbons and oxygenates, while the poor selectivity limits its further use^{2,3}. C₂₊ products (e.g., C₂H₄, C₂H₆, ethanol, and propanol) generally have higher energy densities and larger

economic values compared to C1 products (CO, CH₄, and formate), and therefore provide a more attractive direction to aim for⁴.

Ethylene and ethanol are of great interest because they are economically and energetically valuable and are produced in substantial quantities globally (> 30 MtC/yr)². Literature has found that these two products share the same generation pathway after C-C coupling^{5,6}, which is believed to be the rate-determining step (RDS) of C₂₊ product formation⁷⁻⁹. Considering that C-C coupling usually requires a high surface CO coverage (i.e. high θ_{*CO}), and that Cu is sluggish in converting CO₂ to *CO, tandem catalysts combining Cu and a CO-selective metal (Ag, Au, and Zn) have shown to be advantageous in C₂₊ product generation^{5,10-22}.

Taking advantage of the high electrical conductivity, eCO₂RR activity, and cost-effectiveness, Cu-Ag bimetallic/alloy catalysts have been widely investigated^{18-20,23-29}. The reported Cu-Ag interaction is strongly dependent on the interplay between Ag deposition content, preparation procedures, and the resulting catalyst structure/morphology, as well as phase miscibility, which has led to different product distributions. For example, Higgins et al. synthesized a CuAg alloy with physical vapor deposition (PVD).²³ Compared to pure Cu, the CuAg selectivity of acetaldehyde increased at the expense of ethanol. This was attributed to the reduced oxophilicity when introducing Ag in the Cu lattice, which weakened the adsorption of acetaldehyde and thus its further reduction to ethanol. A similar phenomenon was also observed on CuAg bimetallic catalysts with separated Cu and Ag phases^{20,30}. Although with distinctive catalyst structures (Ag nanoparticle-decorated Cu nanocubes³⁰ and galvanically exchanged Cu foil with Ag²⁰), Clark et al. and Herzog et al. both attributed the weakened carbonyl-containing intermediate adsorption to the locally compressive strain by introducing Ag in the Cu surface. In contrast, Su et al. suggested an improved oxygen affinity for oxygen-adsorbed intermediates instead, resulting in

improved ethanol production when the Cu/Cu₂O composite was dispersed with trace amounts of Ag³¹.

The above divergencies demonstrate a yet-to-be-fully-understood mechanism of the interaction between Cu and Ag in eCO₂RR, which generally falls into three aspects: i) if Cu and Ag have miscible phases, ii) if there is electronic and/or structural interaction between Cu and Ag, and if so, iii) how is the intermediate adsorption influenced and the resultant variation in selectivity. Being able to track the catalysts' variations in crystallite and coordination environment in the eCO₂RR environment can provide more insightful information regarding the above points. Therefore, *operando* measurements are indispensable in helping support traditional H-cell fundamental work.

Herein, CuAg multiphase alloys with various Ag concentration were prepared by magnetron co-sputtering, which not only ensures Cu/Ag interfaces, but also produces a consistent deposition that rules out potential inconsistencies in facet distributions or morphology. Additionally, compared to conventional PVD (used in Ref ²³, magnetron sputtering also improves the deposition efficiency and homogeneity. This approach allows for the focus to be on the intrinsic performance of the Cu-Ag interaction. The result presented herein showed eCO₂RR performance with an improved selectivity towards carbonyls at the expense of H₂ and hydrocarbons upon Ag introduction. To provide further insight, the structural and electronic effects between Cu and Ag were investigated by *operando* grazing incident X-ray diffraction (GIXRD) and grazing-incident X-ray absorption spectroscopy (GIXAS). Combined with density functional theory (DFT) calculations, these analyses lead us to conclude the altered product propensity was attributed to the shared electron cloud density redistribution and strain effect due to Ag incorporation, which lowers the d-band center of Cu and therefore weakens the adsorption of CO and other oxygen-

containing intermediates. Although several previous studies on similar systems have proposed a variety of possible Cu-Ag interaction mechanisms, this work evidenced these hypotheses by *operando* X-ray measurements combined with computational simulations to make the argument more solid.

RESULTS AND DISCUSSION

Catalyst preparation and *ex situ* characterizations

All catalysts were co-deposited on a Si wafer by magnetron sputtering, which not only enables a controllable composition, but has also been shown to yield metastable miscibility between this two thermodynamically immiscible metals²⁴. In such a way, the Cu/Ag interfaces are also ensured, which minimized the influence of geometric and structural engineering and therefore enables focusing on the intrinsic performance of the CuAg system.

The near-surface Cu/Ag composition of each sputtered sample was examined by X-ray photoelectron spectroscopy (XPS). Figure 1 shows the *ex situ* XPS spectra of all as-prepared catalysts with various Ag concentrations. No impurity element was found in the survey scans (Figure S1) of as-prepared and post-mortem catalysts. Low Energy Ion Scattering (LEIS) spectra of as-prepared (Figure S2a) and post-mortem (Figure S2b) samples both exhibit an absence of impurity elements. Two satellite peaks in Cu 2p orbital (Figure 1a) indicate the existence of Cu²⁺, which is inevitable for *ex situ* XPS measurements considering that Cu is easily oxidized once exposed to air. The weaker peak intensity of Ag 3d scans (Figure 1b) after eCO₂RR compared to their as-prepared counterparts appeared in all compositions suggests a less concentrated Ag surface composition after eCO₂RR. Near-surface composition (5-10 nm) of all samples before and after eCO₂RR are listed in Table S1. It is worth pointing out that since Cu

and Ag are both susceptible to oxygen in the air, information on the oxidation state and near-surface element composition obtained from *ex situ* XPS can only provide a preliminary evaluation, while *operando* X-ray measurements (*vide infra*) will be able to give better insight into the catalyst in its active state. Note that although the real Cu-Ag composition is slightly different than expected, the nominal composition was used in this paper for convenience and simplification.

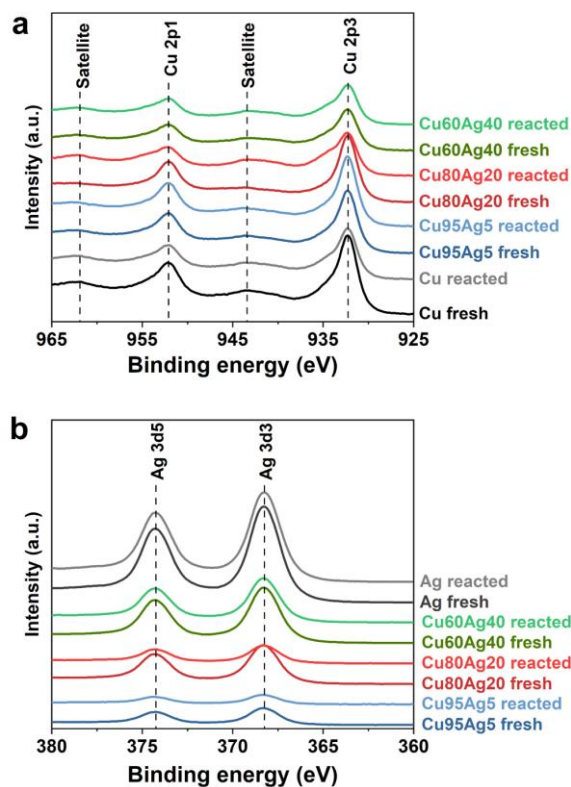


Figure 1. *Ex situ* high-resolution XPS element scan spectra of samples with various Ag compositions: a) Cu 2p orbital and b) Ag 3d orbital.

To understand the influence of Ag incorporation on the crystal structure of Cu, *ex situ* grazing incident (GI)XRD patterns (Figure 2) of as-prepared samples were collected. Both pure Cu and Ag samples agree well with the characteristic Cu and Ag peaks without noticeable position

shifting. After Ag was introduced into the Cu, two broad peaks appeared at a 2θ of 35-45°, where peak A (blue area) and peak B (green area) most likely stem from the shifted and broadened Ag (111) and Cu (111) peaks, respectively. Peak shifting indicates lattice strain of the Cu and Ag phases induced by each other^{32,33}. Additionally, the reduced signal-to-noise ratio and broadened peaks may be attributed to the worse atomic arrangement and smaller crystallite size, respectively, as a result of inter-diffused Cu and Ag domains.

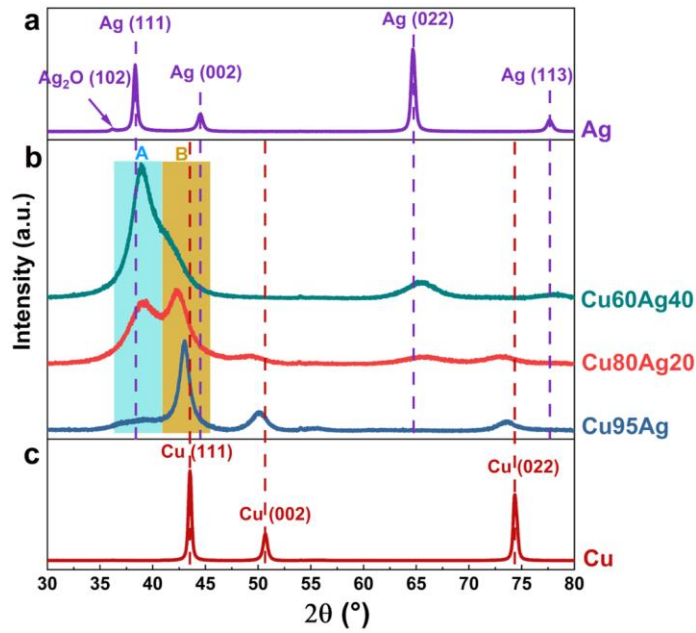


Figure 2. *Ex situ* XRD pattern of a) Ag, b) $\text{Cu}_x\text{Ag}_{100-x}$, and c) Cu. Dash lines indicate the peak position of each facet in the pure metal samples. Highlighted A and B areas indicate the Ag- and Cu-rich phase, respectively.

Thus, these XRD results suggest the formation of multiphase alloys in the CuAg samples, where one is Cu-rich (with Ag being the minority), and the other is Ag-rich (with Cu being the minority). This conclusion is supported by the DFT-based convex hull, shown in Figure S15. No uniform phases of CuAg with fixed composition could be identified by theoretically sampling

varying CuAg compositions and arrangements, suggesting a phase separation proficiency of the two elements. Nevertheless, since it is the Cu sites that produces multicarbon products, the Cu-rich phase is expected to be dominating. Therefore, our following analyses will be mainly focusing on the Cu-rich phase (however both phases will still be considered for H₂ and CO production as explained below).

The absence of peaks related to oxidized Cu in the pure Cu sample indicates that the incident beam penetrated much deeper than the topmost sample layers. Although *ex situ* XRD measurements were conducted in a grazing incident geometry, the incident angle could only be as shallow as 0.7° beyond which the signal-to-noise ratio was insufficiently high for an accurate analysis.

Scanning electron microscopic (SEM) images (Figure S3) also depict smaller particle size of the Cu₈₀Ag₂₀ sample (Figure S3b) compared to that of pure Cu (Figure S3a) and pure Ag (Figure S3b), which agrees with the broadened peaks in *ex situ* XRD results. Energy-dispersive X-ray spectroscopic (EDS) mapping (Figure S4) exhibits a uniform elemental distribution on the Cu₈₀Ag₂₀ sample.

Electrochemical CO₂ reduction performance

Figure 3a gives an overview of the distribution of all detected products for Cu, Ag and various Cu-Ag alloys as reflected by their Faradaic Efficiency (FE). (Selectivity regarding the produced number of moles of each detected product was plotted in Figure S6.) In contrast to Cu being able to produce a variety of products, Ag mainly produces CO, as well as a small amount of H₂ and formate under eCO₂RR conditions. The slightly above 100% total FE may be due either to an experimental error in measuring or not using outlet flow rates from the reactor due to

experimental difficulties. Either way, this slight offset is not expected to affect the trends and conclusions of this work.

Figure 3b shows a promoted CO while suppressed H₂ production, as the Ag concentration increased. This is as expected given that Cu is known to have a higher H₂ selectivity, whereas Ag has a strong selectivity towards CO. In CuAg alloys, CO produced on Ag can also create a high local CO partial pressure at the adjacent Cu sites^{14,18,20}, leading to a CO-enriched local environment, which facilitates the following C-C coupling and thus suppresses direct *CO hydrogenation to produce CH₄, as shown in Figure 4b and c. Previous DFT calculations also revealed weakened *H binding on the Cu surface as *CO coverage increases³² with experimental results validating this³³. Considering that Cu is located at the weak binding side of the volcano plot of H₂ evolution, a weaker *H binding energy will further suppress H₂ production³⁴. This may be one reason for the hampered production of H₂ (Figure 3b)^{20,35}.

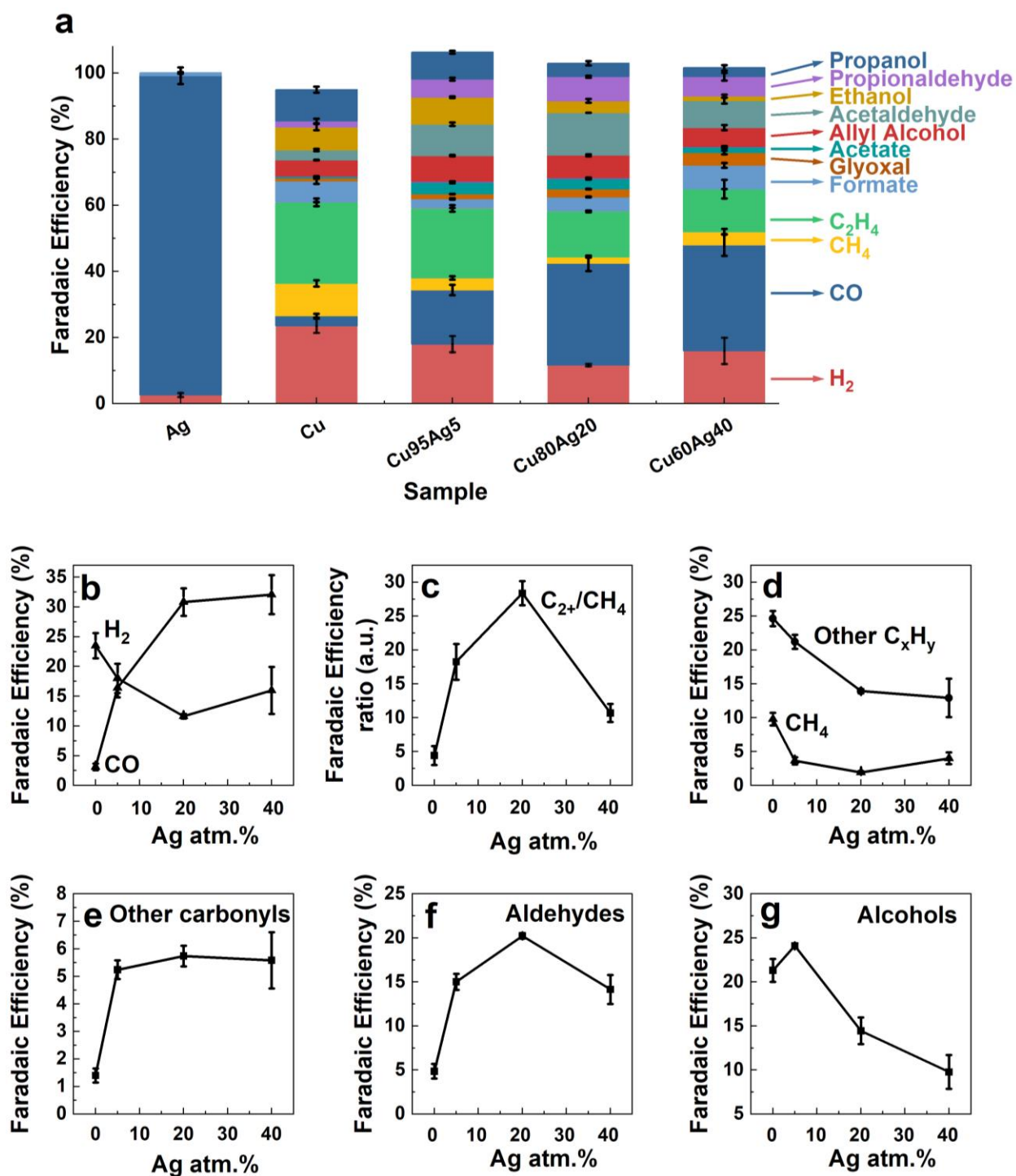


Figure 3. Electrochemical CO_2 reduction performance test conducted in an H-cell with CO_2 -saturated 0.1 M $KHCO_3$ aqueous solution at -1 V vs. RHE for 2 hours. The electrolyte was continuously purged with CO_2 at 10 sccm during e CO_2 RR. a) Faradaic Efficiency of all detected

products. b) Faradaic Efficiency of H₂ and CO as a function of Ag content, and c) Faradaic Efficiency ratio between C₂₊ products and CH₄ (all are *CO-originated) as a function of Ag content. d) Faradaic Efficiency of CH₄ and other hydrocarbons (CH₄, C₂H₄, C₂H₆, C₃H₆, and C₃H₈) as a function of Ag content. e)-g) Faradaic efficiencies of *CO-originated products containing C-O bond as a function of Ag content: e) other carbonyls (glyoxal, acetate, and hydroxy acetone; formate is not included), f) aldehydes (glycolaldehyde, acetaldehyde, and propionaldehyde), g) alcohols (ally alcohol, ethanol, and propanol).

Aldehydes and other carbonyls (excluding formate since it branches from CO and other *CO-originated products in an early stage²) both exhibit a selectivity peak with Cu₈₀Ag₂₀ (Figure 3e and f), accompanied by restrained alcohol (Figure 3g) and hydrocarbon (Figure 3d) production. Introducing Ag would be expected to lead to a local compressive strain of the Cu lattice, due to the larger atomic size of Ag ($r \sim 172$ pm) compared to Cu ($r \sim 128$ pm). This strain effect should shift the Cu valence band structure to deeper levels, which reduces the binding energy of intermediates^{36,37} as well as the oxophilicity (the tendency to form a chemical bond with oxygen atoms of intermediate molecules) of Cu, leading to weakened adsorption of carbonyl intermediates on the Cu surface^{20,38,39}. Another theory that has been proposed to explain the reduced oxophilicity of Cu is that electron density shifts from Cu to Ag at the interface, driven by the difference in their electrochemical potential^{15,40}. The above electronic interaction would decrease the electron density of Cu^{15,39,41,42} at a negative potential^{30,43,44}. As a result, aldehyde would desorb from the catalyst surface before being further reduced to alcohols^{20,30,44}.

To gain more insights into the effects of Ag on C-C coupling, Figure 3c looks at the branching point between C-C coupling to C₂₊ products versus protonation to the C1 product CH₄. The promoted C₂₊/CH₄ ratio primarily occurs by the suppressed CH₄ formation upon alloying with

Ag, as shown in Figure 4b. This can be a convolution of multiple reasons: i) modified adsorption strength of $^*\text{CO}$ versus $^*\text{H}$ as mentioned above (because previous studies have found that the rate-limiting step for methane production is involved with $^*\text{H}$ ⁴⁵), ii) preferential Ag deposition on undercoordinated sites versus terraces, and iii) Ag tends to agglomerate into small islands due to its lower surface energy⁴⁶. A prominent decline of the $\text{C}_{2+}/\text{CH}_4$ ratio occurs with alloys with Ag content greater than 20 atm.%. This could be attributed to that as Ag composition continuously increases, the excessive Ag atoms will cover too many Cu sites, leaving less Cu-Cu neighbors where CO^* adsorption and C-C coupling happen. Considering that C-C coupling requires two adsorbed CO^* in adjacent while CH_4 formation only requires one, a lower $\text{C}_{2+}/\text{CH}_4$ ratio is expected. This phenomenon has been observed by scanning electron microscope (SEM) on CuAg thin films under eCO₂RR conditions, where the importance of an optimal CO^* availability was emphasized for an efficient C-C coupling⁴⁰.

Current density (at a constant potential) as a function of the near-surface Ag composition (as measured *via* XPS) enables investigating the reaction kinetics influenced by Ag incorporation. This is plotted in Figure 4. The dashed lines are linear predictions by assuming that the Cu and Ag sites function separately without interaction between them. The lower total current density relative to the linear expectation shown in Figure 4a indicates a reduced overall activity upon Ag introduction. DFT calculations on the segregation energy of Ag on various Cu facets indicate the propensity of introduced Ag to distribute on the Cu under-coordinated sites compared to terraces (Figure S14). Considering that literature has widely shown that under-coordinated sites are generally more active than terraces in polycrystalline Cu^{2,47-49}, the overall activity is expected to be reduced. Moreover, previous studies have found that by occupying Cu undercoordinated sites with Ag, not only was $^*\text{H}$ binding energy weakened and thus suppressed HER (as explained in

Figure 3b)²³, but further reduction of acetaldehyde to ethanol would also be mitigated²⁶. With Ag preferentially depositing on the undercoordinated sites, this effect can also influence the C₂+ versus methane product ratio.

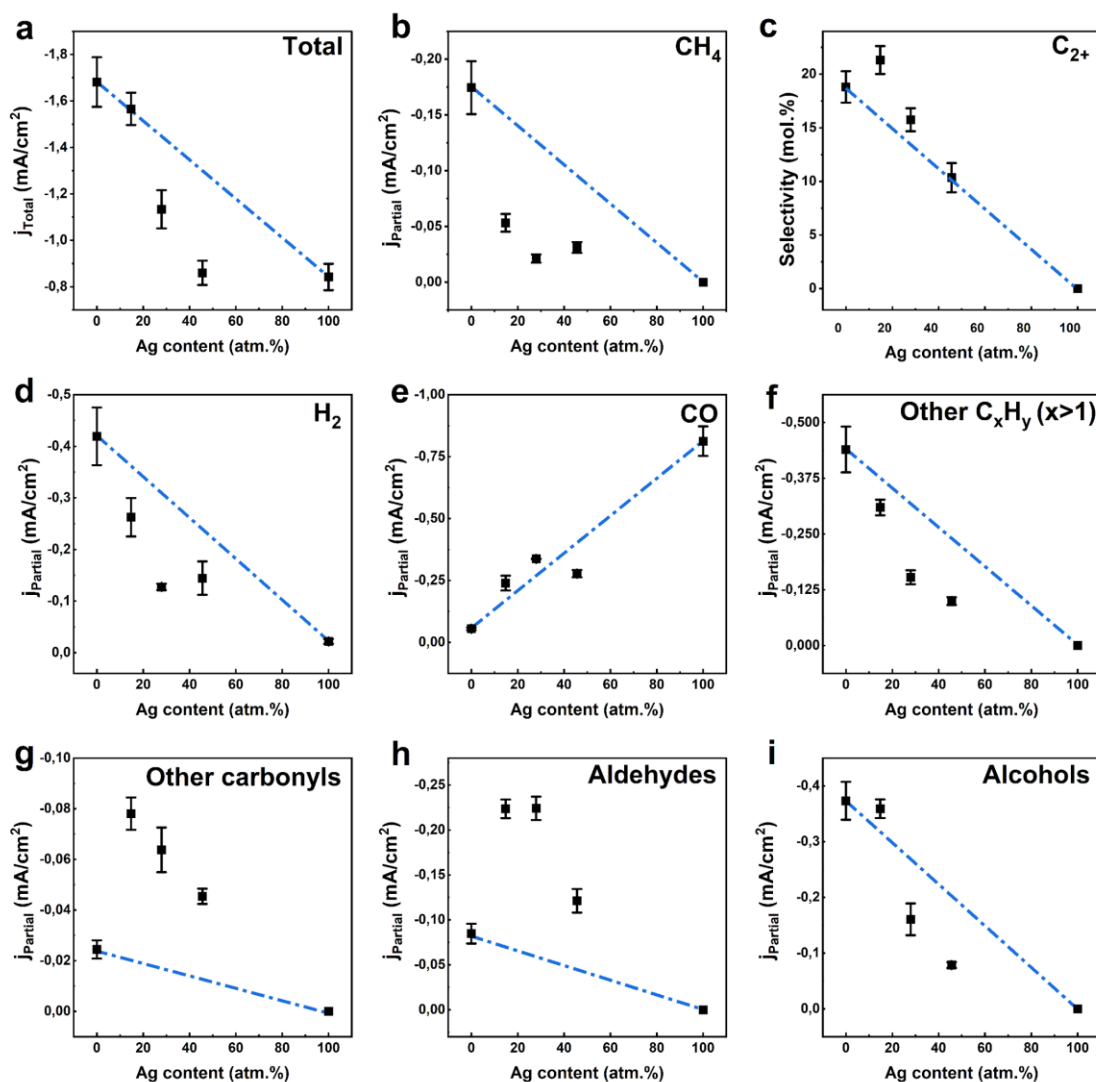


Figure 4. Current density as a function of Ag content. a) Total current density. Partial current density of b) CH₄, c) C₂⁺ products, d) H₂, e) CO, f) other C_xH_y excluding CH₄, g) other carbonyls excluding aldehydes, h) aldehydes, and i) alcohols.

The good agreement of Figure 4 (partial current density) and Figure 3 (Faradaic Efficiency) of each product classification reveals a synergistic effect between Cu and the incorporated Ag on the facilitated aldehydes/other carbonyls production at an expense of H_2/C_xH_y formation.

From the above discussion, the altered product preference could be a convolution of several effects created by introducing Ag in Cu, including spatial effects, compressive strain, oxophilicity variations, electronic interaction between Cu and Ag, CO^* and *H adsorption, Ag incorporation site, etc. To deconvolute these possibilities, *operando* X-ray measurements were employed to provide structural and electronic information on the Cu-Ag interaction in the course of eCO₂RR. Considering that the most promising performance (i.e. the most promoted aldehydes/other carbonyls generation and facilitated C-C coupling) occurs on sample Cu₈₀Ag₂₀, this alloy was used in the following synchrotron measurements.

***Operando* grazing incident X-ray diffraction (GIXRD)**

To monitor the crystal structure of the catalyst during eCO₂RR, *operando* GIXRD was conducted. Given that catalysis occurs at the surface, the surface sensitivity of the measurement geometry was analyzed with pure Cu initially. Since Cu is known to be easily oxidized when exposed to air, the top surface is expected to resemble an oxidized Cu (CuO_x). Figure 5a exhibits a prominent peak related to CuO at open circuit voltage (OCV), which is in stark contrast to its absence at -1 V vs. RHE. Oxidized Cu being reduced to metallic Cu under eCO₂RR conditions have been widely observed by SERS.^{49–51} The poor signal-to-charge ratio is a consequence of the fast scan rate due to beamtime limitations. Nevertheless, the disappearance of the prominent CuO(211) peak at negative potential further validated the surface sensitivity of the incident angle of 0.15° (this is much shallower than what was used in *ex situ* XRD since the *operando* X-ray beam source energy is higher), corresponding to ~2.8 nm of penetration depth (~13.3

monolayers). The corresponding X-ray attenuation depth and penetrated monolayer numbers of each sample were listed in Table S2 (in the case of the Cu₈₀Ag₂₀ alloy, the density of each phase was estimated to be a linear molar combination of each element). In parallel, the crystal size of all samples was estimated *via* the Scherrer equation listed in Table S3. X-ray penetration depth was estimated by assuming linear combination of Cu and Ag atoms according to their molar fraction (Table S4). Results show that the X-ray beam penetration depth was below the value of the corresponding crystal size, demonstrating that it penetrates a homogeneous crystal structure, and therefore further validates the surface sensitivity of GI X-ray measurements. It is worth noting that since our samples are not atomically flat, the estimated X-ray penetration depth could be deeper than calculated. However, the disappearance of the oxidized Cu shown in Figure 5a well confirms the surface sensitivity qualitatively.

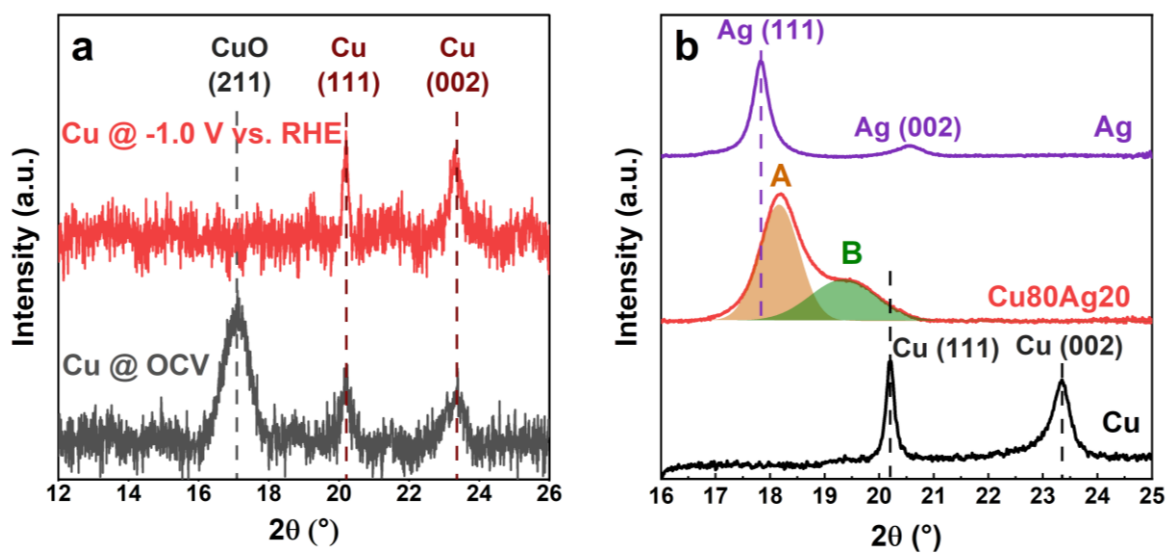


Figure 5. a) Operando GIXRD surface sensitivity verification on Cu at the incident angle of 0.15°. The broad peak between 2θ 16-18° in the OCV scan (black) is assigned to the phase resulting from the mixture of CuO (0 0 2) and (1 1 -1), and its disappearance at -1.0 V vs. RHE (red) indicates the absence of oxidized Cu, which validates the surface sensitivity. **b) Operando**

GIXRD pattern of Ag (top), Cu₈₀Ag₂₀ (middle), and Cu (bottom), respectively. Dashed lines indicate the peak position of corresponding facets in pure metal samples. Dashed lines indicate the corresponding peak position in the pure Cu and Ag samples. Highlighted area A and B indicate the deconvoluted Ag- and Cu-rich phase in the Cu₈₀Ag₂₀ sample, respectively. Backgrounds are removed.

The different peak position from that in *ex situ* XRD is induced by the applied X-ray source energy. Table S5 demonstrates peak matching between *ex situ* (8.04 keV) and *operando* (17 keV) GIXRD measurements.

During *operando* measurements, a working electrode potential of -1 V vs. RHE was applied to the catalyst and kept for 1 hour, during which 6 consecutive scans (~10 min for each scan) were collected and averaged to reduce the signal/noise ratio. Figure 5b shows the same phenomenon regarding peak shift as seen via *ex situ* measurements (Figure 2).

***Operando* grazing incident X-ray absorption spectroscopy (GIXAS)**

An XAS spectrum is comprised of two regions: the X-ray absorption near edge structure (XANES) region, which is located within ~100 eV around the absorption edge; and the extended X-ray absorption fine structure (EXAFS) region, which extends up to ~1000 eV after the absorption edge. The XANES spectra identify the oxidation state and local geometric structure of the absorber atom; while the EXAFS region contains information on the local chemical environment surrounding the central absorber atom, such as coordination number, atomic shell distance, etc.⁵²

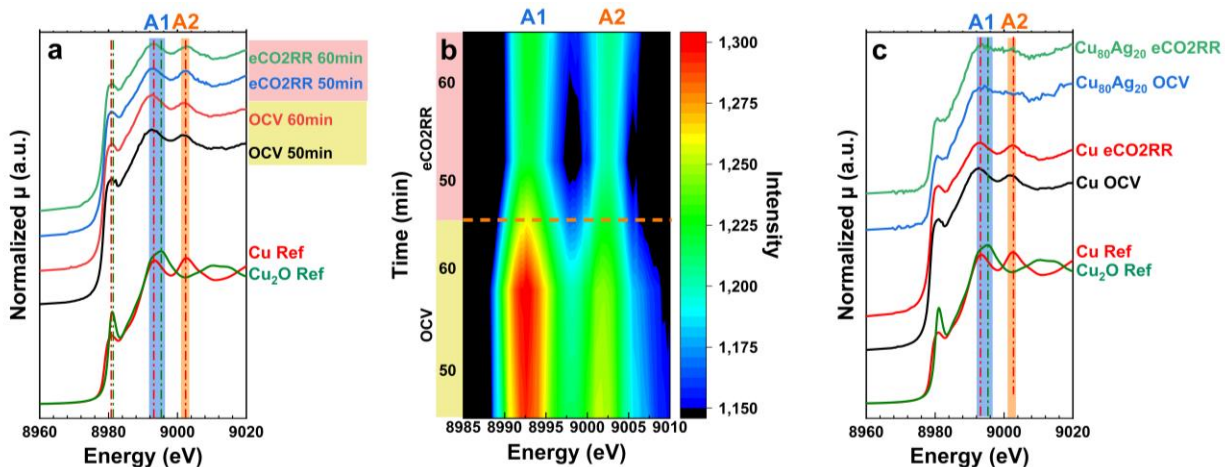


Figure 6. Operando XAS scans at the Cu K-edge. a) XANES region of Cu and Cu₂O references (bottom), and the pure Cu sample at open circuit voltage (OCV) and -1 V vs. RHE (eCO₂RR), respectively. The labels 50 min and 60 min mean the amount of time the samples had been exposed to the corresponding conditions. b) Heat map of the highlighted energy ranges A1 and A2 in a) within 8985-9010 eV. c) Comparison of Cu and Cu₈₀Ag₂₀ samples at OCV and COR conditions, respectively. Each spectrum was the average of two scans at 50 and 60 min being exposed to the corresponding condition, respectively. Spectra of the Cu and Cu₂O references are plotted at the bottom. Each spectrum was subtracted by its pre-edge background and normalized by the edge height.

The surface sensitivity of GIXAS was verified by comparing XANES spectra taken at different incident angles. In order to produce a reasonable signal-to-charge ratio, an incident angle of 0.25° corresponding to a penetration depth of ~2.9 nm (13.9 monolayers) was used for the following GIXAS measurements. The sensitivity test is similar to that in the case of GIXRD. Further details can be found in Figure S7 and Table S6. At the bottom of Figure 6a, both pure Cu and Cu₂O reference XANES spectra are shown. While the absorption edge (8980-8982 eV) has a subtle difference, there is a notable difference in the energy range of 8990-9020 eV under open

circuit voltage (OCV) and eCO₂RR (-1 V vs. RHE) conditions: metallic Cu has one peak in the energy range of 8991-8997 eV (noted as A1) and one peak at ~9002 eV (noted as A2), with the same intensity; while Cu₂O only exhibits one peak in A1. Since XANES gives integral information of all detected Cu atoms regardless of oxidation state, a combination of metallic and oxidized Cu would have a higher intensity of A1 relative to A2, since both oxidation states contribute to the former while only metallic Cu contributes to the latter. Therefore, the more distorted intensity of A1 to A2 is, the more Cu₂O there is. This enables a qualitative indication of the transformation from Cu₂O to Cu at the surface.

The top four scans in Figure 6a show the pure Cu sample at OCV and subsequently being exposed to eCO₂RR conditions (@ -1.0 V vs. RHE). The spectra at OCV appear to be primarily metallic Cu, which is expected given that the penetration depth of 2.9 nm is a bit thicker than the native oxide layer, entailing a strong background Cu signal. However, a detailed look at the data showed there is a noticeable difference. To make the spectra difference more explicit, Figure 6b enlarges the energy range of 8985-9010 eV and transforms it into a heat map plot. It can be clearly seen that A1 has a higher intensity than A2 at OCV while both areas reached the same intensity under eCO₂RR conditions.

Figure 6c compared pure Cu to the Cu₈₀Ag₂₀ sample, where an evident difference can be seen. The worse signal-to-noise ratio of the Cu₈₀Ag₂₀ spectra can be attributed to the lower Cu composition as well as smaller crystallite sizes and/or worse crystallinity, as also indicated by the XRD results (Figure 5). Nevertheless, the raw XANES profile without background subtraction or signal intensity normalization (Figure S9) exhibited the same trend as in Figure 6. In contrast to pure Cu, where two separated peaks in A1 and A2 are present, only one wide wave in the energy range of 8985-9010 eV exists in the Cu₈₀Ag₂₀ spectra. Thus, these spectra appear to be more

similar to the Cu₂O reference. This suggests a reduced electron density in the Cu phase upon introducing Ag, compared to the case of pure Cu. Furthermore, this reduced electron density would also suggest that less electron density variation would be seen when switching from open circuit voltage to eCO₂RR conditions. As expected, a zoom-in of the Cu₈₀Ag₂₀ during the transition from OCV to eCO₂RR similar to Figure 6b shows a much less noticeable variation in A₁ relative to A₂ in the corresponding energy range (Figure S8). XANES spectra at the Ag L₃-edge were also analyzed but did not show significant changes as Ag composition varied (Figure S13). Note that any factor that changes the electron density of the absorbing atom could influence the XANES profile shape, including but not limited to oxidation state, electronegativity, as well as the number and identity of ligand(s) and binding atom(s). In order to build a concrete conclusion from the above observed trends, DFT calculations were also employed later in this work.

Computational simulations

In order to verify the strength of the found charge transfer and strain effects and their influence on the binding of eCO₂RR reaction intermediates, we performed DFT calculations on the CuAg system. Although our XRD results indicate the existence of two phases, one Cu-rich and the other Ag-rich (Figures 2 and 5), considering that both the nominal and near-surface composition (Table S1, Figure S1, and S2) indicates the dominating composition of Cu in general, and that it is the Cu sites that are active for C₂⁺ product formation, our simulation focused on the Cu-rich phase. Both the bulk and surface structures of Cu with varying Ag content (x_{Ag}) were screened, where for each x_{Ag} all symmetrically unique atomic arrangements were sampled and the most stable motifs were identified (see Methods section for further details). Note that only metallic Cu

was considered in the simulations based on the experimentally observed disappearance of CuO_x under eCO₂RR conditions.

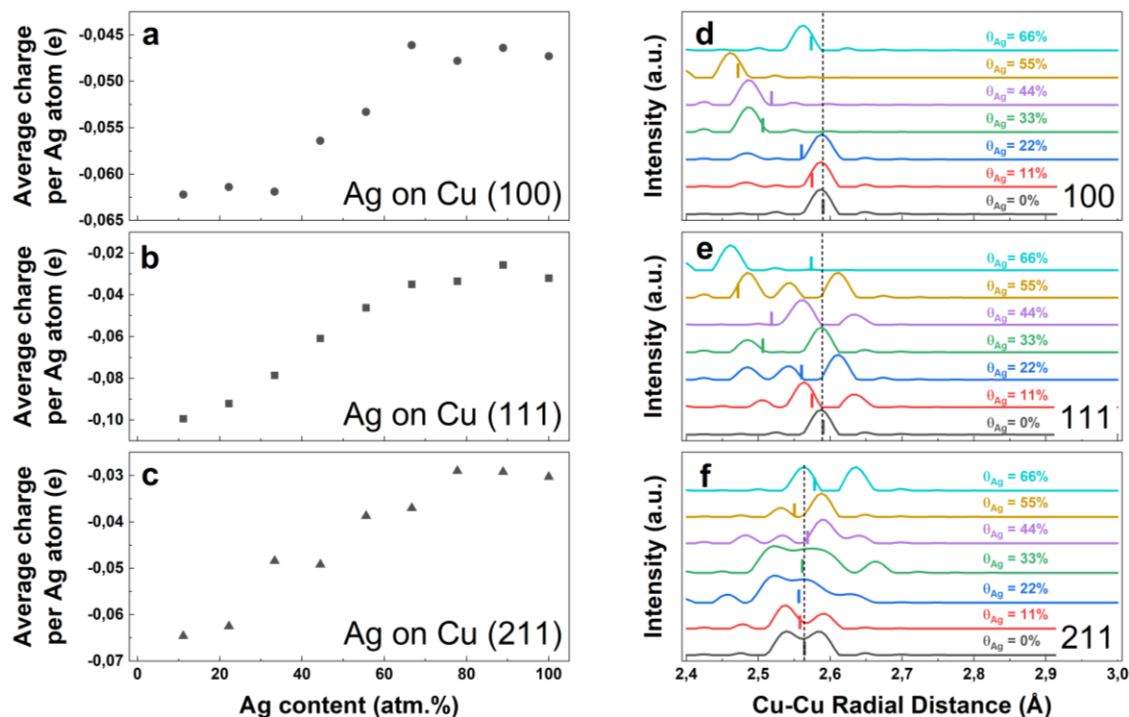


Figure 7. DFT calculated charge transfer and Cu-Cu distances on $\text{Cu}_x\text{Ag}_{100-x}$ surfaces. a-c)

Average partial charges per Ag atom in $\text{Cu}_x\text{Ag}_{100-x}$ when Ag atoms are introduced on Cu (100) (a), (111) (b), and (211) (c) facets, respectively. d-f) Cu-Cu radial distance distribution in $\text{Cu}_x\text{Ag}_{100-x}$ when Ag atoms are introduced to Cu (100) (d), Cu (111) (e), and Cu (211) (f), respectively. Averaged Cu-Cu bond length of each Ag content on each facet is indicated by a bar with the same color. Dashed lines represent the Cu-Cu bond length without Ag introduction.

The performed DFT calculations support the spectroscopic suggestions of an electron density shifting in the Cu phase upon Ag introduction compared to pure Cu. In the case of dilute Ag concentrations in the bulk crystal (< 30 atm.%) where Ag atoms are generally surrounded by Cu, each Ag atom gains *ca.* 0.14 electrons on average, leading to a charge redistribution in the

surrounding Cu atoms in order to keep a neutral charge in the crystal (Figure S17). The same qualitative behavior was found for Ag covering the Cu crystal surface, as shown in Figure 7a-c for the (111), (100) and (211) facets of Cu with varying Ag surface coverage. However, the magnitude of the charge on Ag is reduced with respect to the bulk crystal as the number of neighboring Cu-atoms is decreased on the surface. The determined charge per Ag atom linearly decreases beyond an Ag coverage of 30%, as Ag atoms start to adjoin each other.

Our calculations indicate that introducing Ag into a Cu bulk crystal leads to an overall increase in the Cu-Cu distances (Figure S18) as a consequence of the crystal lattice expanding linearly with the Ag content (Figure S16), reflecting the GIXRD measurements (Figure 5b). In contrast, introducing Ag atoms on the Cu surface leads to a net compressive strain among surface Cu atoms, as shown in Figure 7d-f. The compressive strain is significant on the terrace facet (100) and (111), where a reduction in the mean Cu-Cu distances up to 0.1 Å with increasing Ag surface coverage is found. In contrast, surface steps, exemplarily simulated as the (211) facet, experience no significant strain. The varying strain on different facets suggests that surface terraces might react more strongly to an introduction of Ag on Cu. On the other hand, we emphasize that Ag tends to preferably segregate on Cu-steps rather than terraces, leading to a more pronounced deactivation of under-coordinated sites for the reaction towards products beyond CO (Figure S14).

In order to identify the magnitude of the respective effects arising from introducing Ag on the surface of Cu-rich domains, we further studied the effect of Ag on eCO₂RR adsorbate binding. We note that although Ag generally tends to segregate to the surface (Figure S14), the binding of *CO on surface-Ag is strongly disfavored compared to Cu-sites. Thus, the latter is likely responsible for the activity and we focus on the ligand and strain effects surface Cu-atoms

experience from neighboring Ag-atoms. The two effects are intertwined and a rigorous splitting of them is, in principle, impossible. In order to distinguish between them, in this work, we define ligand effects as the effect on adsorbate binding upon exchanging Cu atoms neighboring the active site with Ag at fixed atomic geometry, while strain effects are referred to the change in binding strength resulting from geometric rearrangement on the catalyst surface. The main focus is to evaluate which effect plays the dominant role in the altered production distributions.

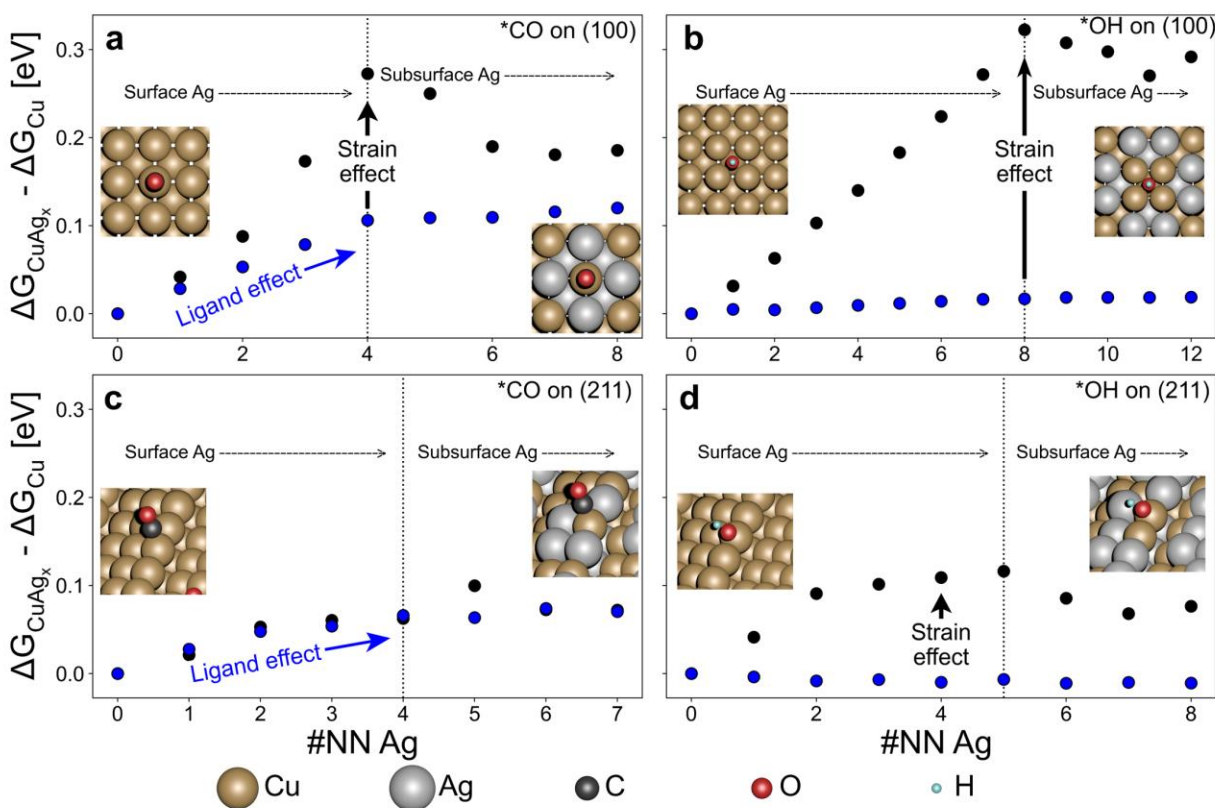


Figure 8. DFT-calculated Gibbs free energy of *CO binding on a) *CO on Cu (100), b) *OH Cu(100), c) *CO on Cu(211), and d) *OH on Cu (211) facets with varying number of nearest neighboring Ag atoms (#NN Ag).

The *CO binding energy reduces upon increasing the number of neighboring Ag atoms (#NNAg) due to a pure ligand effect, as shown in blue in Figures 8a and c. Interestingly, the

reduction in binding strength only occurs when in-plane neighbors of the *CO-adsorbing Cu atom are substituted with Ag, whereas introducing Ag to the subsurface layer did not have a significant effect. We attribute this reduced effect in the subsurface to a higher saturation of Ag in the subsurface, reducing the charge transfer from specific neighbors. Atomic relaxation leads to a distancing of the Cu-atoms from Ag, while at the same time reducing the distance to the neighboring Cu-atoms. This created compressive strain adds to the ligand effect in the reduction of *CO binding strength, as indicated by the black arrow in Figure 8a and c. In the case where Cu is completely surrounded by Ag on the surface, the effect of strain maximizes. Upon further increasing the number of Ag neighbors, the Cu atom is lifted slightly above the original surface plane, increasing its distance to all neighbors and the second layer (distance to second layer increased up to ~ 0.3 Å). This weakened coordination strength of Cu to the neighboring atoms enhances *CO binding again.

Figure 8 depicts the binding energy of *CO (a,c) and *OH (b,d), on Cu (100) and (211) facets, respectively. *CO is a descriptor for eCO₂RR activity, while *OH binding can be used as a descriptor for the binding of oxygen-bound species (*OR), as verified by the scaling of *OCH₂CH₃ with *OH in Figure S20. The Cu(100) surface terrace and (211) step were chosen, as the former is often identified as the most active facet for CO-dimerization^{53,54}, while surface steps are generally considered as most relevant for the adsorption of weak binding reaction intermediates⁵⁵, such as acetaldehyde. The nearest neighbors (NN) to the Cu-atoms binding the corresponding intermediates were chosen as the Ag deposition sites. All possible Ag arrangements were sampled but only the most stable CuAg surface structures were considered in Figure 8. In all four considered scenarios, Ag energetically favored substituting all neighboring surface Cu-atoms before substituting the subsurface.

In order to first simulate the pure ligand effect, plotted as blue dots in Figure 8, as defined above, we first introduced Ag on Cu while keeping the atomic positions frozen, i.e., the introduced forces resulting from substituting Cu with Ag were not allowed to relax. Subsequently, the atoms were allowed to relax the introduced forces, giving rise to a strain effect, as presented in black dots in Figure 8.

Independent of the adsorbate and surface facet, adsorption strength is reduced upon introducing Ag, a consequence of a downshift in the d-band center (Figure S21). Furthermore, the effect of introducing Ag is strongly enhanced on the (100) terrace (Figure 8 a-b) compared to the (211) step sites (Figure 8 c-d).

In contrast to *CO, the *OH binding strength responds barely to a pure ligand effect. Note that while *CO tends to adsorb in a top site, we identified the binding site of *OH as the fourfold-hollow site on the (100) facet (Figure 8b) and in a bridge site on the (211) step (Figure 8d).

These binding motifs make *OH much more sensitive to strain effects compared to *CO.

Considering that carbonyl products, especially aldehydes, bind on Cu with *O before desorbing or being further reduced, the large reduction in binding strength due to strain explains their promoted production.

It has been found that both a compressive strain and the slightly reduced electron density in the Cu sites, as demonstrated by XANES results, could shift the pristine Cu d-band center to a deeper level and hence weaken the intermediate adsorption^{36–39}, which coincides with the attenuated Cu-C*/O* shell (peak B) in EXAFS spectra on the Cu K-edge shown in Figure S10. More details and analysis regarding EXAFS data can be found in Supporting Information Section 3.3.

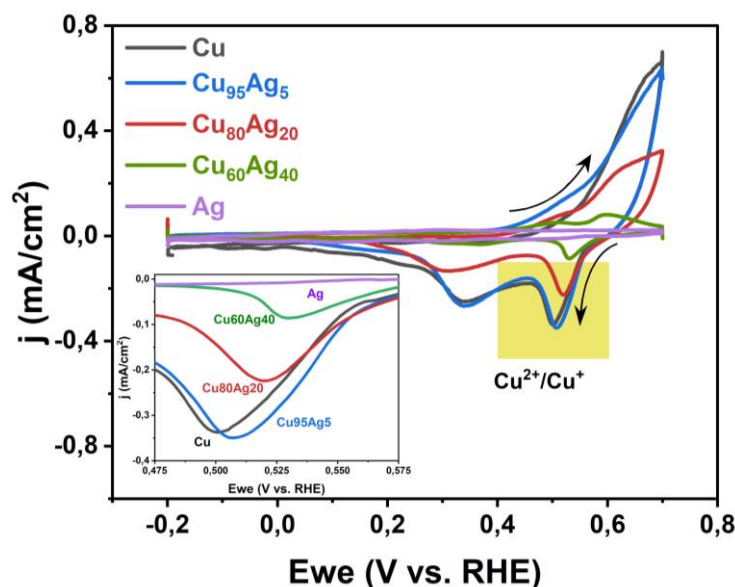


Figure 9. Cyclic voltammetry scans for determining oxophilicity of Cu from Ag incorporation in CO₂-saturated 0.1 M KHCO₃ at the scan rate of 50 mV/s. Inset is the enlarged highlighted area of the Cu²⁺/Cu⁺ wave.

The compressive strain was also found to reduce the oxophilicity of transient metals^{31,56,57} and thus weaken the adsorption of species binding on the metal surface with O* (e.g., aldehydes and other carbonyls). It has been found that the oxophilicity of transition metal surfaces is highly correlated with their standard reduction/oxidation potential²⁰. Therefore, cyclic voltammetry scans in the potential window between -0.2 to 0.7 V vs. RHE were performed to investigate the redox properties (Figure 9). As expected, an anodic shift of the Cu²⁺/Cu⁺ reduction wave is observed, confirming a reduced oxophilicity upon the introduction of Ag in Cu²⁰. This is further evidence to help explain the promoted production of aldehydes and other carbonyls observed in Figure 3. It is worth pointing out that for Cu catalysts, it is often the Cu²⁺/Cu⁺ onset potential is compared to evaluate the catalysts' oxophilicity. However, onset potential of bimetallic catalysts involves with kinetic effects. Specifically, an earlier onset potential occurs not only due to a

lower oxophilicity, but also with a higher surface Cu composition. In this regard, CV feature position is more straightforward.

CONCLUSIONS

In this work, we prepared a series of CuAg multiphase alloys with varying Ag content for a better understanding of product selectivity resulting from electrochemical CO₂ reduction. Among the alloys, Cu₈₀Ag₂₀ (nominal composition) exhibited the highest selectivity towards C-C coupling and carbonyl production. *Operando* GIXRD confirmed the spatial strain effect between Cu and Ag phases. The Cu K-edge XANES suggests a slight electron density decrease in the Cu phase upon Ag introduction, compared to pure Cu. DFT calculations demonstrated both the electronic interaction and strain effect of Cu upon Ag incorporation, which shifted the Cu d-band center to a deeper level and thus weakened intermediate adsorption. The oxophilicity of Cu was also reduced as a consequence of compressive strain. Taking together, the facilitated generation of multicarbon aldehydes and carbonyls at the expense of H₂ and hydrocarbon production can be attributed to a combined effect of a deeper d-band center and reduced oxophilicity as Ag is introduced. This work provides insights into the interaction between alloyed metals during electrochemical CO₂ reduction on an atomic scale, which can be used to direct new bi-/multi-metallic/alloy designs in this field.

ASSOCIATED CONTENT

Supporting Information. Experimental and simulation methods, details of element distribution acquired from *ex situ* XPS, sample configuration of *operando* XAS measurements on the Ag L₃-edge, LEIS spectra and peak assignment, surface sensitivity verification and penetration depth of *operando* X-ray measurements, *operando* XRD and *ex situ* XRD 2 θ match, DFT-calculated

facet-dependent Ag surface segregation energies, DFT-calculated convex-hull, DFT-calculated lattice expansion with Ag content, Bader charges on Cu and Ag in the bulk crystal, calculated Cu-Cu radial distances in the bulk crystal, calculated effect of pure strain on the binding of *CO and *OH, scaling of the binding energy of *OCH₂CH₃ with the binding strength of *OH, d-band center shift with the #NN_{Ag}.

This information is available free of charge on the ACS Publications website.

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Notes

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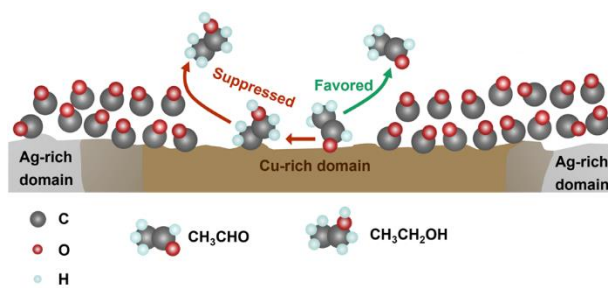
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Supporting Information for

Mechanistic Insights into Aldehyde Production from Electrochemical CO₂ Reduction on CuAg Alloy via Operando X-Ray Measurements

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1. THIN FILM CATALYST FABRICATION AND CHARACTERIZATION

Cu_xAg_{100-x} (X = 100, 95, 80 and 60, nominal content) were fabricated with a high vacuum (base pressure < 1 × 10⁻⁷ mbar) magnetron sputtering chamber (AJA International, Inc.) using direct current (DC) mode. The Cu and Ag targets (Kurt–Lesker, 99.9999%) were (co)-deposited on 75 nm thick Si wafers (Wafer World Inc., boron-doped, resistivity < 0.025 Ohm-cm) or Kapton foil (DuPont, only for XAS at Ag L3-edge) as the substrate. In both cases, the substrate was sputter-cleaned by Ar at 40 W for ~5min to remove any surface contaminations and oxide layers. A ~5 nm thick Ti adhesive layer was then sputtered at a rate of ~0.7 Å/s, followed by a ~50 nm thick Cu_xAg_{100-x} catalyst layer deposited at a rate of ~1.0 Å/s. The deposition rate and nominal composition as well as the thickness of the deposited catalyst layer were calibrated by a quartz crystal microbalance (QCM). All samples were deposited under a 5 mTorr, 10 sccm Ar (N5, Air Liquide) atmosphere at room temperature.

X-ray photoelectron spectroscopy (XPS) was performed by a ThermoScientific Thetaprobe instrument equipped with an Al K α X-ray source. The operating base pressure was kept below 1.0 × 10⁻¹⁰ mbar. Survey and high-resolution elemental scans were recorded with a step size of 1.0 eV and 0.1 eV, respectively; while in all cases pass energy of 200 eV and dwell time of 50 ms were used. The data were acquired and analyzed with Thermo Advantage (Thermo Fisher Scientific). Low energy ion scattering (LEIS) was processed on the same instrument, utilizing a focused He ion gun (1 keV) at a base pressure of 2.0 × 10⁻⁷ mbar. The LEIS spectra were acquired with a step size of 1.0 eV.

X-ray diffraction (XRD) was conducted on a Malvern Panalytical Empyrean X-ray diffractometer in the grazing incident geometry with the incident beam angle at 0.7°. An Empyrean Cu LFF HR gun (K α_1 = 1.540598 Å, 8.04 keV) was used as the X-ray source and operated at 45 kV and 40 mA.

Scanning electron microscopy (SEM) measurements were performed on a FEI Quanta 200 FEG instrument at an accelerating voltage of 10 kV.

Table S1 Element composition acquired from ex situ XPS. Three spots were taken on each sample and the average is listed in the table.

Sample (nominal composition)	Element Distribution										
	As-prepared					After CO2ER					
	Cu	Ag	Cu in CuAg	O	C	Cu	Ag	Cu in CuAg	O	C	K
Cu	37.0	0	100.0	42.2	20.8	26.1	0	100.0	42.8	31.1	0
Cu ₉₅ Ag ₅	33.4	5.8	85.2	38.1	22.7	36.5	2.9	92.6	42.29	17.4	1.5
Cu ₈₀ Ag ₂₀	36.4	14.1	72.1	28.7	20.8	28.4	7.4	79.4	41	23.17	0
Cu ₆₀ Ag ₄₀	24.9	20.8	54.4	28.3	26.0	25.2	14.9	62.8	34.6	25.0	0
Ag	0	49.2	0.0	13.5	37.3	0.0	44.8	0.00	13.6	41.6	0

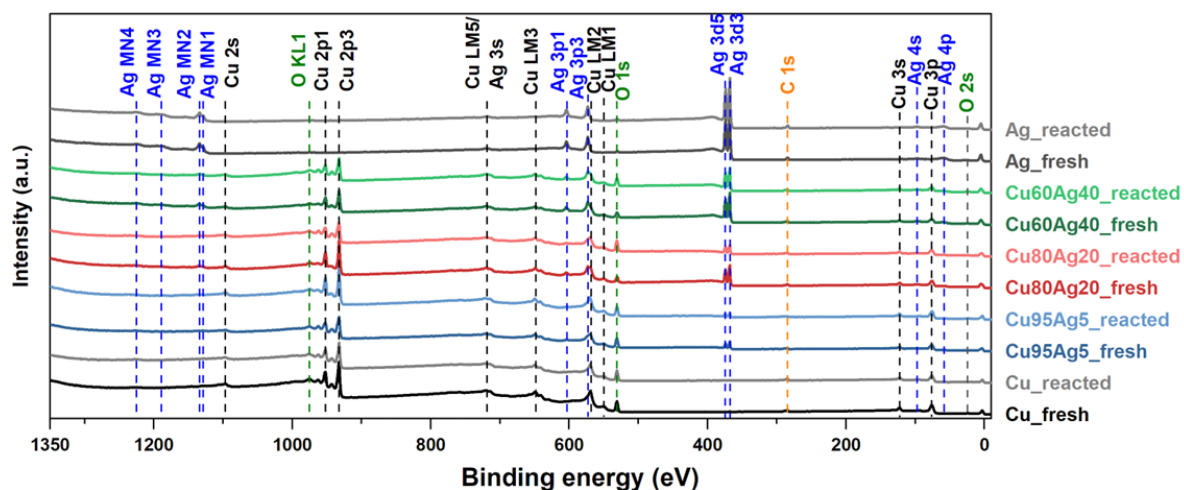


Figure S1 XPS survey of samples with varying Ag contents.

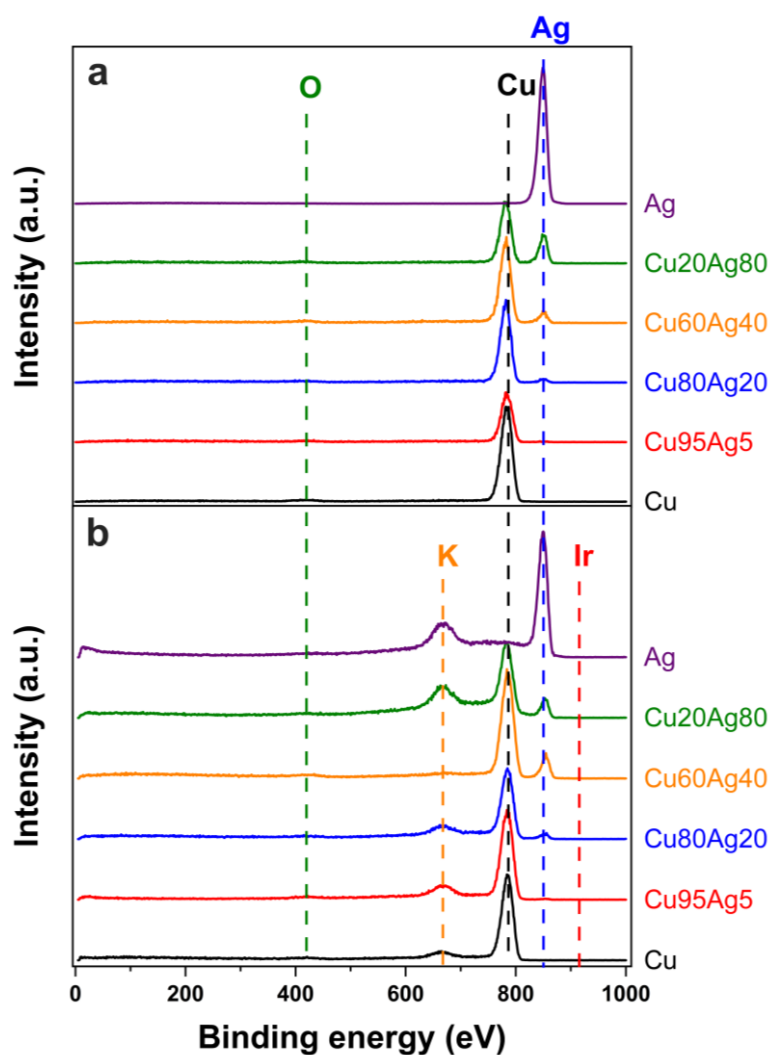


Figure S2 LEIS survey of samples with various Ag concentrations. (a) Samples before eCO₂RR. Spectra were taken without any pre-treatment of the samples. O was from the oxidation of Cu and Ag, considering that the samples were stored in the atmosphere. No impurity elements were detected. (b) Samples after eCO₂RR for 2 hours. The sample surface was carefully rinsed with MilliQ water and dried with pressed air before the test. Dash lines indicate the peak position of the corresponding element. K was from the not fully removed electrolyte after eCO₂RR. No signal related to Ir was found in any sample.

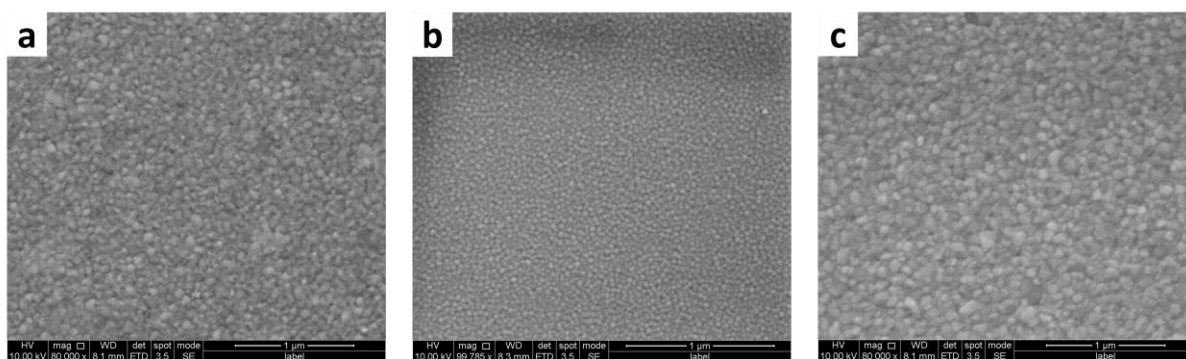


Figure S3 SEM images of a) pure Cu, b) Cu₈₀Ag₂₀, and c) pure Ag.

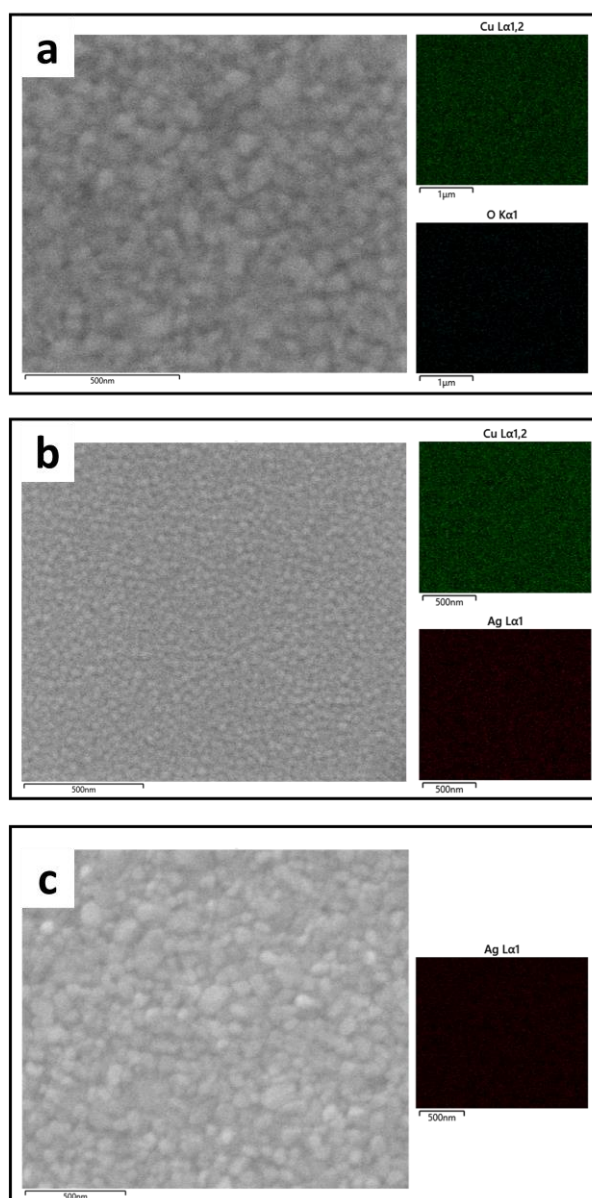


Figure S4 EDS elemental mapping images of a) pure Cu, where O comes from oxidized Cu; b) Cu₈₀Ag₂₀, and c) Ag.

2. ELECTROCHEMICAL CO₂ REDUCTION MEASUREMENTS

K₂CO₃ solutions were prepared by dissolving K₂CO₃ (99.995% trace metals basis, Aldrich) into MilliQ water (18.2 MΩ.cm @25°C, 2ppb TOC, Q-POD®). 0.1M KHCO₃ solution was prepared by purging CO₂ (N45, Air Liquide) into 0.05 M K₂CO₃ solution for at least 30min, until pH~6.8.

Electrochemical CO₂ reduction was carried out in a 2-chamber H-cell in the 3-electrode configuration. The two chambers were separated by a hydrocarbon-based ion-exchange membrane (Selemin AMV, AGC Engineering CO., LTD.), activated in 0.1 M KOH (99.995% trace metals basis, Sigma-Aldrich) overnight, and stored in MilliQ water. The liquid volume of both chambers is 6.5 mL, whereas the actual electrolyte volume was 6.0 mL in both chambers. The cathode chamber was purged with CO₂ at 10 sccm for the duration of the experiment. An IrO₂-coated carbon paper (1.0 cm x 1.0 cm, Dioxide Materials) and a miniature leak-free Ag/AgCl/Saturated KCl electrode (Innovative Instruments Inc.) were used as the counter and reference electrode, respectively.

A stagnant H-cell was chosen because the current is more stable without electrolyte flow. Moreover, it allows for concentrating on the intrinsic performance of the catalysts by minimizing potential influence from unstable liquid flow and/or dead volumes, as well as varied mass transfer conditions in different runs.

Electrochemical experiments were controlled using a BioLogic SP-200 potentiostat. Unless otherwise stated, the working electrode potential (E_{WE}) was recorded and referenced against the Ag/AgCl reference electrode (RE), which was then converted and reported in terms of the reversible hydrogen electrode (RHE) scale following the equation below. The RE was calibrated following the procedure described in ¹ before use.

$$E_{RHE} = E_{WE} + E_{RE} + 0.059 \times \text{pH} \quad (1)$$

Prior to each measurement, a 10 min chronoamperometry (CA) at -0.2 V vs. RHE was applied to reduce the CuO_x layer, which was naturally formed when metallic Cu (or the Cu phase in CuAg bimetallic catalysts) was exposed to air. This potential was chosen because it is more negative than the CuO_x reduction potential (0.35 V vs. RHE) while not sufficiently reductive to produce detectable products. Cyclic voltammetry (CV) was then scanned in the potential window of -0.2 to +0.7 V three times. The redox features were used to primarily check the consistency of the same sample in different runs. A constant potential of -1.0 V vs. RHE was then applied and kept for 2 hours. Gas products were analyzed *in situ* with gas chromatography (GC, Thermo-Scientific, TRACE 1300), while liquid products were collected at the end of the experiment and analyzed with high-performance liquid chromatography (HPLC, Agilent 1200 series). eCO₂RR measurements were conducted three times for each sample.

Figure S3 is a screenshot of the HPLC data processing software (since the diode array detector (DAD) was used for detecting acetaldehyde, only DAD signals are shown here). Insert is enlarged area highlighting acetaldehyde. Different products are loaded in highlighted area. Four spectra are shown here: 0.1 M KHCO₃ electrolyte (red), post-mortem electrolyte after CO₂R reaction on Cu (purple), Cu₈₀Ag₂₀ (yellow), and Cu₆₀Ag₄₀ (green). It gives an example to show the sensitivity of our HPLC equipment in detecting acetaldehyde. It can be seen that the signal-to-noise ratio is sufficient even at

the lowest acetaldehyde concentration (*i.e.*, on pure Cu, read line) for accurately investigating product selectivity.

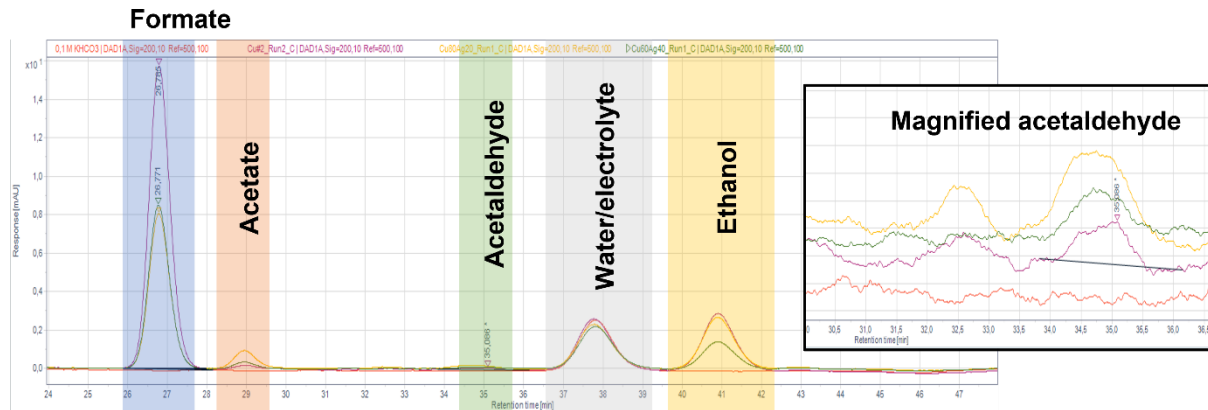


Figure S5 Screenshot of the HPLC data processing software. Insert is enlarged area highlighting acetaldehyde. Different products are labeled in highlighted area. Four spectra are shown here: 0.1 M KHCO₃ electrolyte (red), post-mortem electrolyte after CO₂R reaction on Cu (purple), Cu₈₀Ag₂₀ (yellow), and Cu₆₀Ag₄₀ (green). The selectivity of each product was determined by its Faradaic efficiency, following Faraday's law:

$$FE_i = \frac{z \times F \times n_i}{I_T \times t} \times 100\% \quad (2)$$

where, z is the number of transferred electrons for producing species i , F is the Faraday's constant (96485 C/mol), n_i is the number of moles of produced species i , I_T is the total current of the reaction, and t is the reaction time.

Specifically, gas products are calculated following the equation:

$$FE_i = \frac{z \times F \times P \times v_{Total}}{I \times R \times T} \times A_{i,GC} \times f_{i,GC} \times 100\% \quad (3)$$

where, P is the atmosphere pressure (101.325 kPa), v_{Total} is the inlet CO₂ gas flow rate (10 sccm), T is room temperature (300 K), $A_{i,GC}$ is the integrated peak area of species i in GC, and $f_{i,GC}$ is the calibration factor of species i in GC. (As the total conversion is small, inlet CO₂ gas flowrate is a reasonable approximation for outlet flow rate in this situation.)

Liquid products are measured by the following equation:

$$FE_i = \frac{z \times F \times V_{Total} \times \frac{A_{i,HPLC}}{f_{i,HPLC}}}{Q} \times 100\% \quad (4)$$

where, V_{Total} is the total volume of the cathode chamber (6 mL), T is room temperature (300 K), $A_{i,HPLC}$ is the integrated peak area of species i in HPLC, and $f_{i,HPLC}$ is the calibration factor of species i in HPLC.

Current densities were based on the geometric area of the working electrode (2.25 cm²). Partial current density of each product was evaluated by the following equation:

$$j_i = j_{\text{Total}} \times \text{FE}_i \quad (5)$$

where, j_{Total} is the total geometric current density.

Productivity S_i of each product with respect to the produced number of moles are calculated in the following way:

$$S_i = \frac{n_i}{\sum_{i=1}^n n_i} \times 100\% \quad (6)$$

For gas products:

$$n_i = \frac{P \times v_{\text{Total}}}{R \times T} \times A_{i,\text{GC}} \times f_{i,\text{GC}} \quad (7)$$

For liquid products:

$$n_i = \frac{V_{\text{Total}} \times \frac{A_{i,\text{HPLC}}}{f_{i,\text{HPLC}}}}{Q} \quad (8)$$

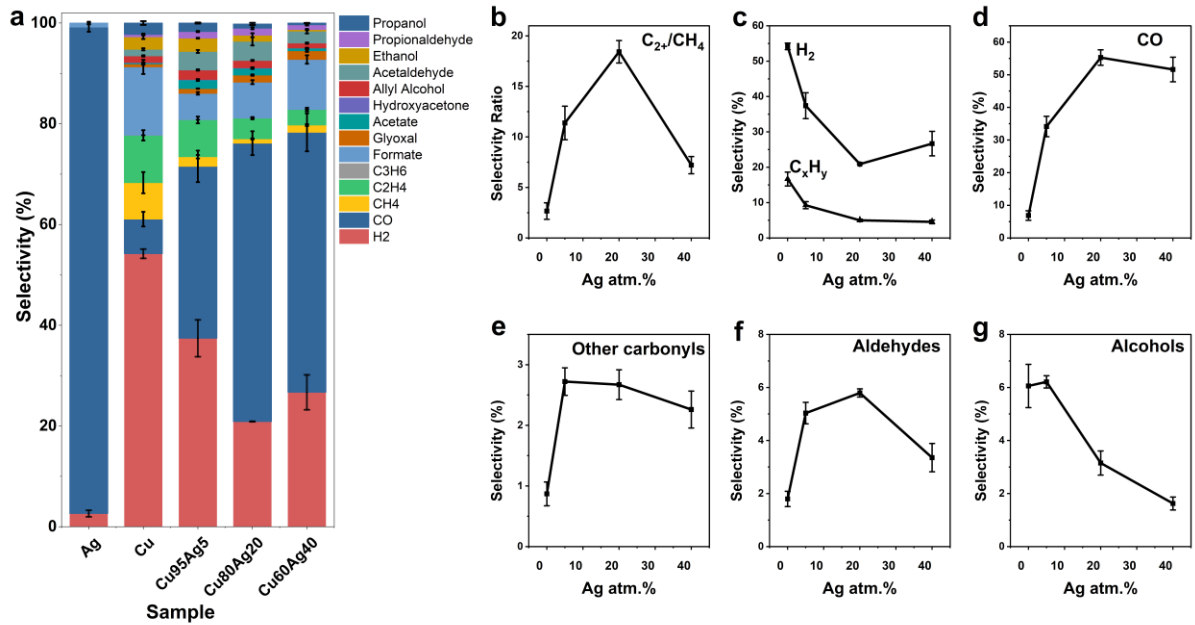


Figure S6 eCO₂RR performance test conducted in an H-cell in CO₂-saturated 0.1 M KHCO₃ aqueous solution at -1 V vs. RHE for 2 hours, plotted in selectivity regarding produced number of moles. CO₂ was continuously purged to the solution at 10 sccm during eCO₂RR. a) Faradaic efficiency of all detected products. b) Faradaic efficiency ratio between *CO-originated products depending on the involvement of C-C coupling during formation process as a function of Ag contents. c) Faradaic efficiency of H₂ and C_xH_y (CH₄, C₂H₄, C₂H₆, C₃H₆, and C₃H₈) as a function of Ag content. d) Faradaic efficiencies of CO₂ as a function of Ag content. e)-g) Faradaic efficiencies of *CO-originated products classified on C-O bond as a function of Ag content: e) other carbonyls (glyoxal, acetate, and hydroxy acetone; formate is not included), f) aldehydes (glycolaldehyde, acetaldehyde, and propionaldehyde), g) alcohols (allyl alcohol, ethanol, and propanol).

3. OPERANDO X-RAY MEASUREMENTS

3.1 Experimental Details

All *operando* X-ray measurements were conducted at the Stanford Synchrotron Radiation Lightsource (SSRL) at ambient pressure and room temperature. Electrochemistry was controlled using a BioLogic SP-200 potentiostat.

Operando GIXRD was conducted at beamline 2-1, using a Si (111) monochromator and beam energy of ~ 17 KeV ($0.729 \text{ \AA}$). *Operando* GIXAS on the Cu K-edge in both X-ray absorption near edge structure (XANES) and the extended X-ray absorption fine structure (EXAFS) regions were collected at beamline 11-2, using a Si (220) LN2-cooled monochromator and a 100-pixel monolithic solid-state Ge detector array, in the transmission mode.

A previously designed 3D-printed flow cell was employed in XRD and XAS at the Cu K-edge (~ 9 keV) to enable the grazing incident configuration of the sample while maintaining a higher horizontal alignment of the sample and also maintaining a high transmissivity for X-ray incidence as described in ². During *operando* XAS, the electrolyte being purged with CO₂ gas was pumped through the cell by a 2-channel peristaltic pump (SHENCHEN LabN6) at 50 mL/min. Cu_xAg_{100-x} catalyst samples were prepared on a Si (111) substrate, as described in Session 2.1. Two 50 μm diameter Pt wires (Alfa Aesar, 99.99%) and a miniature Ag/AgCl electrode (Innovative Instruments Inc.) served as the counter and reference electrode, respectively.

XAS data analysis was processed using SixPack156 ³, Athena and Artemis software ⁴.

3.2 X-ray Diffraction (XRD)

Table S2 **Structural property and elemental composition of samples in operando XRD measurements at $E = 17$ keV ($\lambda = 0.0927$ nm).**

Sample	2θ (°)	Lattice spacing (Å)	Lattice constant (Å)	Molar composition of Cu (%)	Molar composition of Ag (%)
Cu	20.18	2.08	3.60	100	0
Ag	17.83	2.35	4.07	0	100
Ag-rich phase in Cu ₈₀ Ag ₂₀	18.16	2.31	4.00	15.85	84.15
Cu-rich phase in Cu ₈₀ Ag ₂₀	19.34	2.17	3.76	67.14	32.86

The elemental molar fraction is calculated according to Vegard's law.

Table S3 X-ray penetration depth calculation in operando XRD measurements: attenuation and penetration depth calculation at $\alpha = 0.15^\circ$, $E = 17$ keV.

Sample	Density (g/cm ³) [#]	Penetration depth (nm) [‡]	Lattice spacing (Å)	Penetrated monolayer numbers
Cu	8.96	2.76	2.08	13.89
Ag	10.49	2.42	2.35	10.77
Ag-rich phase in Cu ₈₀ Ag ₂₀	10.25	2.23	2.31	9.65
Cu-rich phase in Cu ₈₀ Ag ₂₀	9.47	2.43	2.17	11.20

[#] The density of each phase in Cu₈₀Ag₂₀ was estimated by assuming a linear molar combination of Cu and Ag.

[‡] Penetration depth was calculated based on Beer-Lambert law, assuming the primary intensity is reduced to $i \sim 1/e$ (i.e., ~37% of the initial value).

Table S4 Estimation on the crystal size in operando XRD measurements.

Sample	2 θ (°)	FWHM width (°)	Crystallite size (nm) [§]
Cu	20.18	0.22	18.15
Ag	17.83	0.69	5.82
Ag-rich phase in Cu ₈₀ Ag ₂₀	18.16	0.87	4.56
Cu-rich phase in Cu ₈₀ Ag ₂₀	19.34	1.44	2.77

[§] Crystal size was calculated by Scherrer's formula. The Scherrer constant was 0.94 by assuming spherical crystallites with cubic symmetry.

Table S5 *XRD peak match between operando and ex situ measurements*

Facet	2 θ (°)	
	Operando (17keV)	Ex situ (8.04 keV)
Ag (1 1 1)	17.88	38.33
Ag (0 0 2)	20.67	44.54
Ag (0 0 2)	29.29	64.56
Cu (1 1 1)	20.23	43.54
Cu (0 0 2)	23.36	50.63
Cu (0 2 2)	33.24	74.34
CuO (2 1 1)	17.01	36.43
CuO (1 1 1)	18.07	38.75
CuO (2 0 0)	18.15	38.93
CuO (1 1 -3)	28.04	61.55

3.3 X-ray Absorption Spectroscopy (XAS) on the Cu K-edge

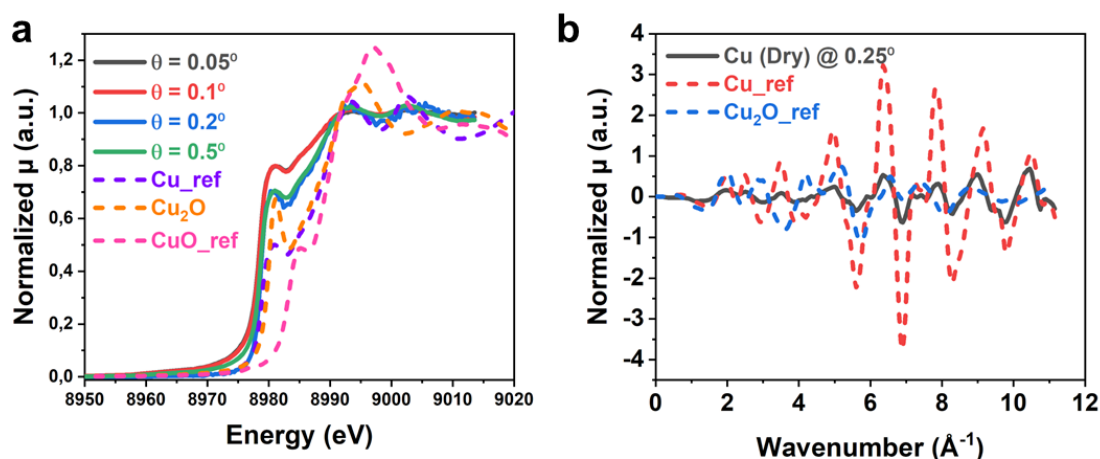


Figure S7 **Operando GIXAS surface sensitivity verification on Cu:** a) XANES region at different grazing incident angles. Solid lines: XANES scans at various incident angle ϑ ; dash lines: reference scans of different Cu oxidation state that were previously taken at the same beamline with the identical setup². The stronger absorption intensity at $\vartheta < 0.2^\circ$ at the edge energy (~ 9 keV) compared to $\vartheta \geq 0.2^\circ$ indicates the more oxidative Cu state. However, a slightly large incident angle is necessary to compensate the weak absorption intensity. Therefore, 0.25° was chosen as the incident angle for the following GIXAS measurements. B) EXAFS plotted in the k space at $\vartheta = 0.25^\circ$. Solid line: scan plotted in the k space at the incident angle $\vartheta = 0.25^\circ$. Dashed lines: reference scans of Cu and Cu_2O .

Table S6 **Operando GIXAS surface sensitivity verification: attenuation and penetration depth calculation at $\alpha = 0.25^\circ$, $E = 9 \text{ keV}$**

Sample	Density (g/cm ³)	Penetration depth (nm) [‡]	Lattice spacing (Å) [£]	Penetrated monolayer numbers
Cu	8.96	2.89	2.08	13.89
Ag	10.49	2.53	2.35	10.77
Ag-rich phase in Cu ₈₀ Ag ₂₀	10.25	2.55	2.31	11.04
Cu-rich phase in Cu ₈₀ Ag ₂₀	9.47	2.77	2.17	12.77

[‡] Penetration depth was calculated based on Beer-Lambert law.

[£] Lattice spacing was calculated from XRD, based on (111) facet for both Cu and Ag, as it is the most prominent phase shown in XRD.

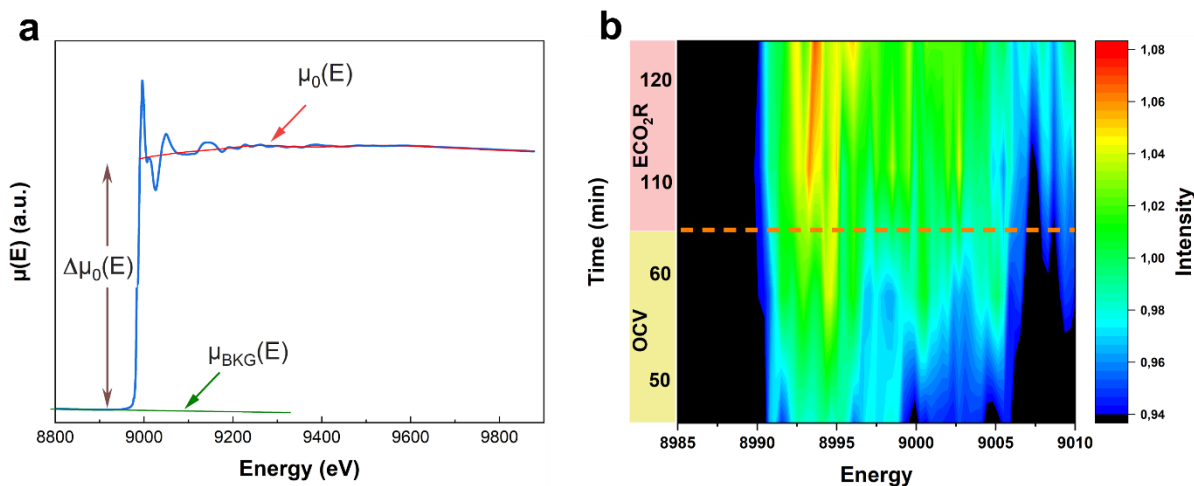


Figure S8 **a) Illustration of XANES data normalization:** the raw data ($\mu(E)$) is extracted by the smooth background ($\mu_{BKG}(E)$), and divided by peak height ($\Delta\mu_0(E)$). This is to eliminate the influence of energy absorbed by other compartments other than the sample itself. In the case of CuAg alloy, the energy can also be absorbed by Ag atoms in the pre-edge region. $\mu_0(E)$ is the smooth “bare-atom” background in the EXAFS fine-structure function and is not for XANES. **b) Heat map of operando XAS scans at the Cu K-edge in the zoomed in XANES region of 8985-9010 eV of the Cu₈₀Ag₂₀ sample.**

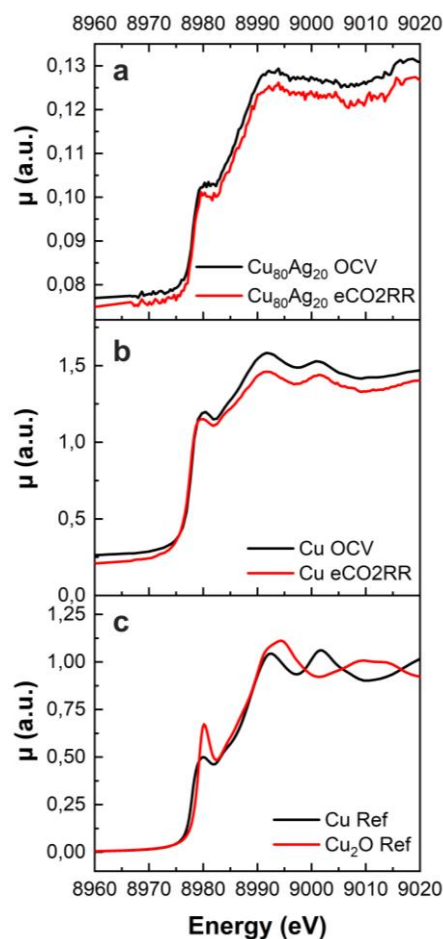


Figure S9 **Raw XANES profile** (i.e., without background extraction or signal intensity normalization) of (a) $\text{Cu}_{80}\text{Ag}_{20}$ at OCV (black) and during eCO_2RR (red), (b) Cu at OCV (black) and during eCO_2RR (red) and (c) Cu reference (black) and Cu_2O reference (red). The XANES signal intensity and noise-to-charge ratio in the $\text{Cu}_{80}\text{Ag}_{20}$ alloy (a) is lower than that of the pure Cu sample (b) in general, which can be attributed to the lower Cu composition and worse crystallinity in the alloy.

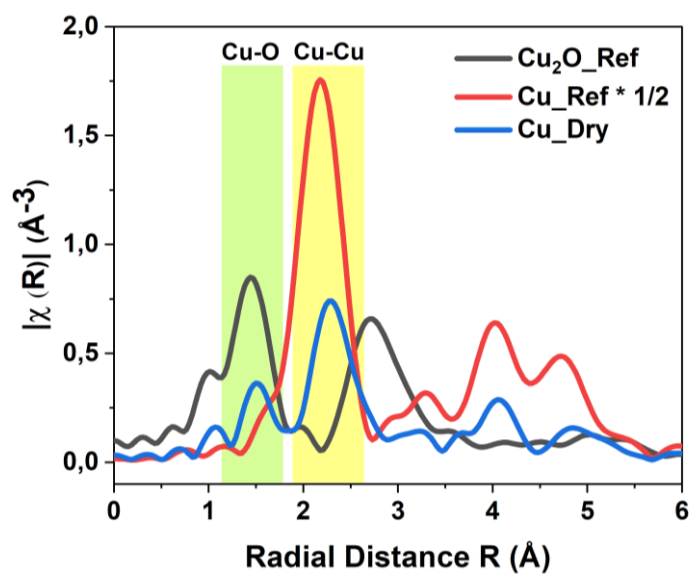


Figure S10 **XAS R space (k^2 weight) peak identification on the Cu K-edge.**

In order to investigate how the electronic interaction contributes to the facilitated acetaldehyde and other carbonyl production, EXAFS measurements on the Cu-edge were Fourier-transformed to radial distance R (i.e., R space) to inspect its local chemical environment. Figure S6 gives a primary assignment of peak positions in the pure Cu sample compared with the metallic and Cu_2O references, where the interatomic distance information surrounding the central atom (i.e., Cu) is given as the R -value. Since atoms in the first shell predominantly contribute to the coordination environment of the central atom^{5,6}, only R values below 3 Å are considered in this work. This is because 3 Å is normally the maximal bond length of the first shell and greater R numbers usually suggest atoms beyond the first shell or multiple scattering⁷. The peak at $R \sim 2.25$ Å is assigned to the first Cu-Cu shell (all R values in this work are phase uncorrected without further notice), while the peak located at ~ 1.6 Å is attributed to the first Cu-O shell, which agrees with the previous studies^{8,9}.

Figures S7b and S7c depicted R space starting after 30 minutes of testing (Cu-30 min) on Cu and $\text{Cu}_{80}\text{Ag}_{20}$, respectively. The observed peaks can be attributed to two main areas contributed by different atomic shells, as compared to Cu and Cu_2O references (Figure 8a). The peak centered at $\sim R$ 1.5 Å is usually assigned to the Cu-O or Cu-C shell, which is normally attributed as the indication of CuO_x ^{5,10}. However, our GIXRD (Figure 5) and XANES (Figure 6a) results have pointed out that Cu was in its metallic state after the first 20 min. Additionally, considering that all intermediates produced during eCO₂RR bind on the Cu surface with either C or O atom(s), and that EXAFS only indicates the existence of an atom at a given distance while fails in atom identification, one potential explanation for peak B is the adsorption of intermediate species. Therefore, the greatly attenuated peak B upon Ag adsorption might suggest a suppressed intermediate adsorption upon Ag incorporation.

The peak centered at $\sim R$ 2.25 Å is normally attributed to the first Cu-Cu or Cu-Ag shell^{8,10,11}. Again, since EXFAS does not identify the neighboring atoms, and considering the close atomic number of Cu and Ag, both shells can contribute to this wide peak. Additionally, considering that XRD results (Figure 5) indicate the co-existence of two phases in the sample, one to be Cu-rich (where Cu is the majority while Ag is the minority) and the other Ag-rich (where Cu is the minority while Ag is the majority), and that EXAFS gives an averaged information for all the detected Cu atoms, the current peak at $\sim R$ 2.25 Å exhibits the interatomic bond distance convoluted with six circumstances upon Ag introduction: i) Cu-Cu right next to Ag in the Cu-rich phase, ii) Cu-Cu outreached the close neighbor of Ag in the Cu-rich phase, iii) Cu-Ag in the Cu-rich phase, iv) Cu-Cu right next to Ag in the Ag-rich phase, v) Cu-Cu outreach the closest neighbor of Ag in the Ag-rich phase, and vi) Cu-Ag in the Ag-rich phase. With such a complicated and convoluted information, it is difficult to draw any concrete conclusion on the strain effect of Ag introduction on the Cu-Cu bond.

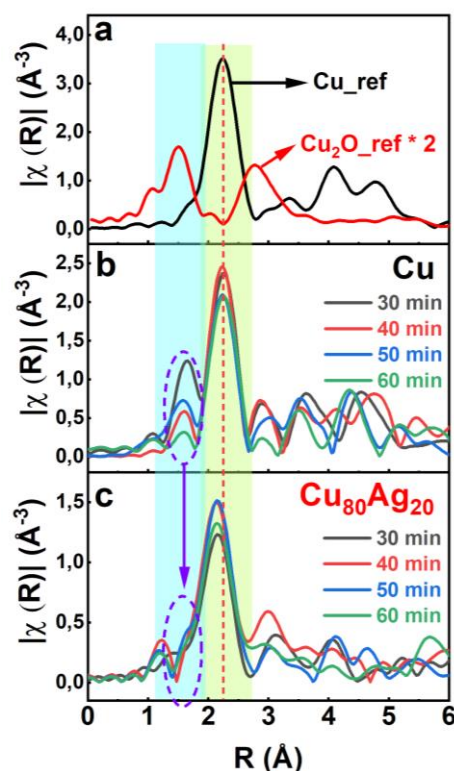


Figure S11 **Fourier-transformed EXAFS R space spectra (k -2 weight) on the Cu K-edge on a) Cu and Cu_2O references, b) pure Cu sample, and c) $\text{Cu}_{80}\text{Ag}_{20}$ sample as a function of time during eCO₂RR.**

Ag L₃-edge (~ 3.4 keV) tender XANES was performed at beamline 4-3 using a LN₂ cooled Si (111) monochromator at $\varphi = 0^\circ$, in the fluorescence mode with a Lytle detector. A custom-designed back illumination bottle cell allowed the X-ray incident to transmit¹². Owing to the soft excitation energy required for Ag L₃-edge, the catalysts were deposited on an ultra-thin 7.5 μm thick Kapton foil to mitigate X-ray attenuation. The layer configuration from the bottom (right next to the substrate) to the top is as the following (Figure S1): i) Ti interlayer (~ 10 nm) providing a better adhesion); ii) Cu sublayer (~ 100 nm) to help maintain the mechanical stability without introducing foreign elements in addition to Cu and Ag during deposition; and iii) the catalyst layer CuAg (~ 5 nm), which is so thin that the incident X-ray beam can easily reach the catalyst/electrolyte interface. The total transmission at the Ag L-edge energy (~ 3.35 keV) was calculated to be above 85%¹³. During the *operando* XANES scans, CO₂ was continuously purged in the electrolyte at 10 sccm. A graphite rod (Ted Pella) and the same miniature Ag/AgCl electrode as abovementioned functioned as the counter and reference electrode, respectively. While XANES experiments were also employed to record the Ag L₃-edge, these variations were quite small and were relatively inconclusive.

3.4 X-ray Absorption Spectroscopy (XAS) on the Ag L-edge

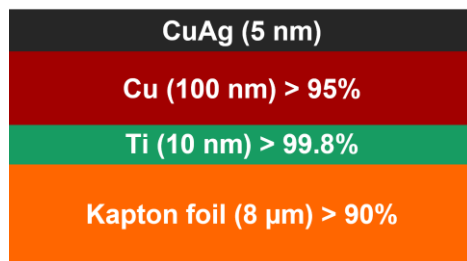


Figure S12 **Sample configuration and X-ray transmission of each layer** of back-scattered bottle cell used for XAS measurements on the Ag L_3 -edge at beamline 4-3

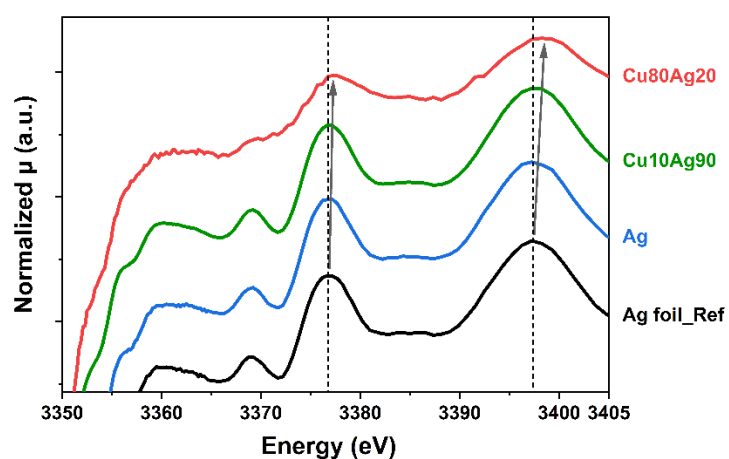


Figure S13 **Operando XAS scans at Ag L_3 -edge in the near-edge region (XANES)** during eCO_2RR of samples with different Ag content, compared with the Ag foil reference, indicated by dashed lines. Arrows emphasize peak position shift.

4. COMPUTATIONAL METHODS

The first principles analyses have been performed by density functional theory (DFT) calculations applying the Vienna ab initio package (VASP) ¹⁴. A plane wave cut-off energy of 500 eV was used throughout, combined with the BEEF-vdW exchange correlation functional ¹⁵. Periodic boundary conditions were applied and the number of k-points were varied based on the respective unit cell sizes. For surface calculations, a vacuum layer of at least 14 Å was added perpendicular to the slab surface, only one k-point was applied in this direction together with a dipole correction to avoid artificial interaction of repeated slabs. The charge distributions in the CuAg compositions were determined via Bader charge decomposition. Both the bulk and surface compositions at varying Cu/Ag ratios were sampled by comparing all symmetrically inequivalent atom distributions and optimizing both the atomic positions and lattice cell parameters. Bulk structures consisting of up to 64 atoms were considered. The most stable structures at each simulated CuAg composition were chosen for the analyses in this article. Segregation energies were calculated as the potential energy difference of a 6-layer thick Cu-slab being doped by Ag on the surface or in the 4th layer.

All theoretical data and analysis routines can be retrieved from https://gitlab.com/gkastlun/cuag_for_c2.git.

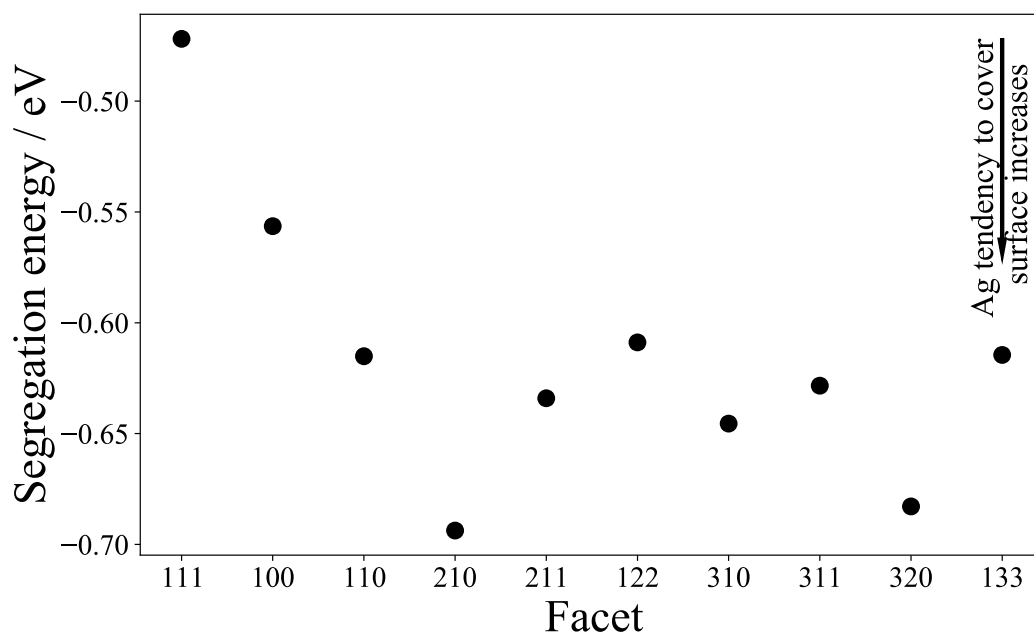


Figure S14 Surface segregation energy of dilute Ag in Cu on a series of surface facets

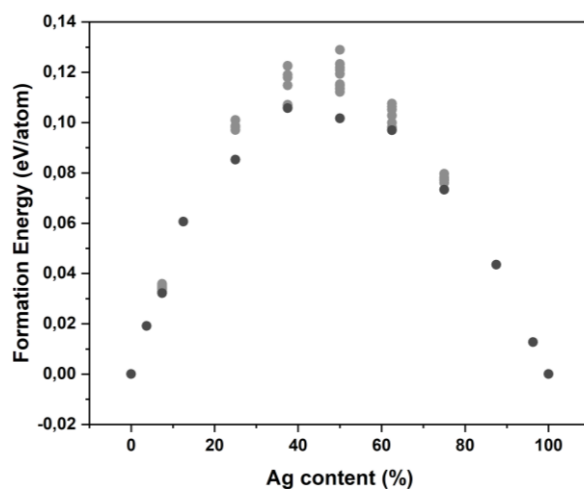


Figure S15 DFT-Calculated formation energies of CuAg crystals with varying composition. All possible CuAg arrangements in 2x2x2 and 3x3x3 unit cells were sampled. No stable phase of a CuAg alloy has been identified

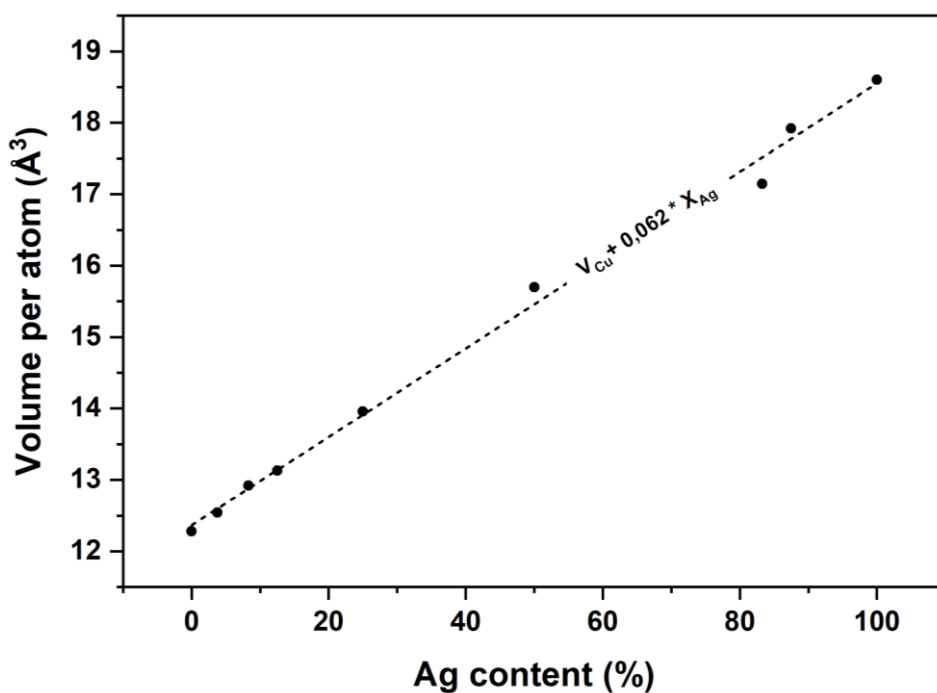


Figure S16 DFT-Calculated volume per atom in a CuAg bulk crystal in dependence of the Ag content. The volume per atom was chosen instead of the lattice constant as varying arrangements of Ag in Cu lead to slight deviations from the cubic symmetry into a tetragonal crystal.

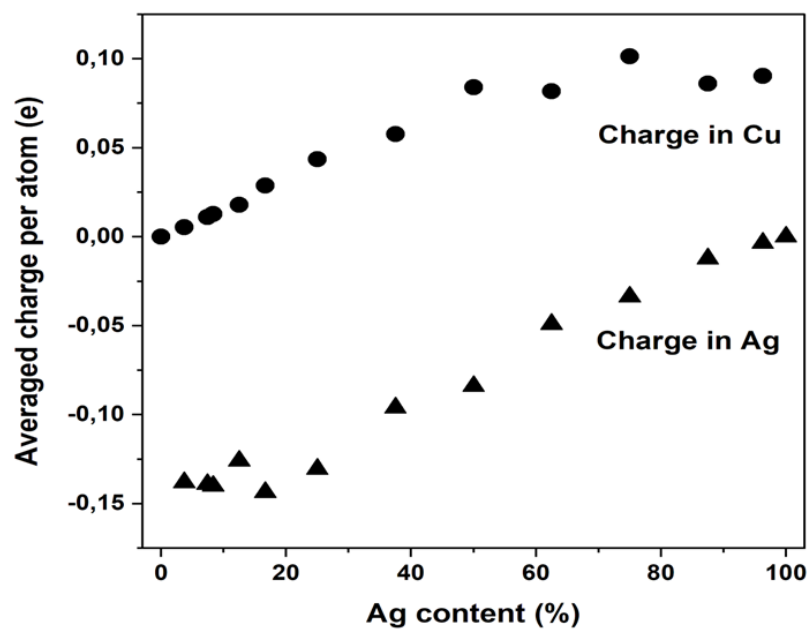


Figure S17 Net mean charge per atom for Cu and Ag atoms in a bulk fcc crystal of varying composition, resulting from Bader charge analysis.

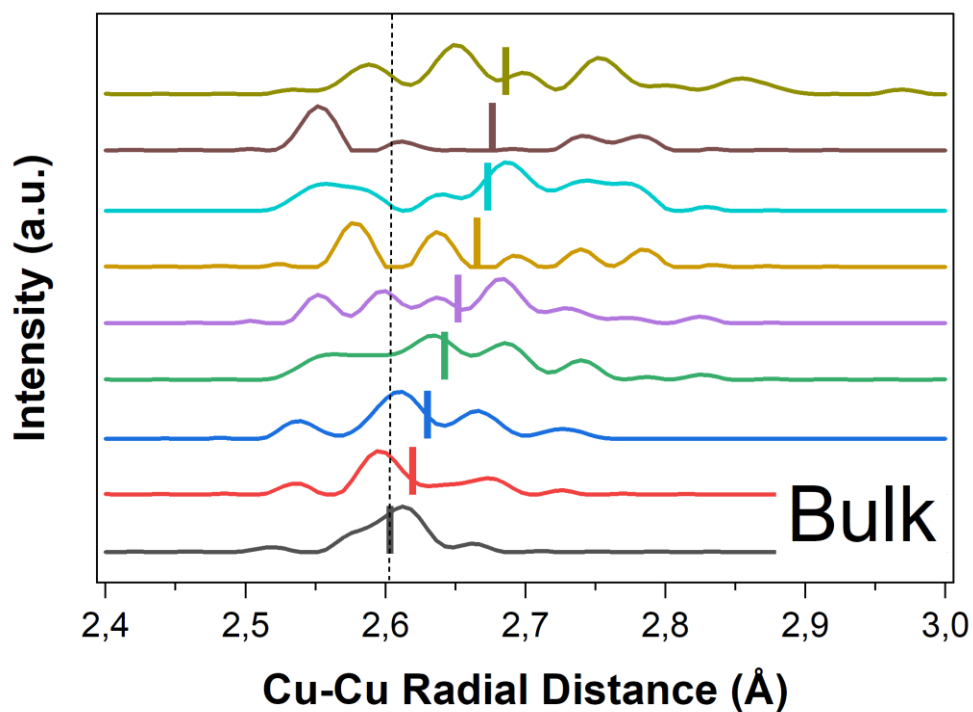


Figure S18 DFT calculated Cu-Cu radial distance in $\text{Cu}_x\text{Ag}_{100-x}$ at varying Ag content right next to the introduced Ag atoms in bulk.

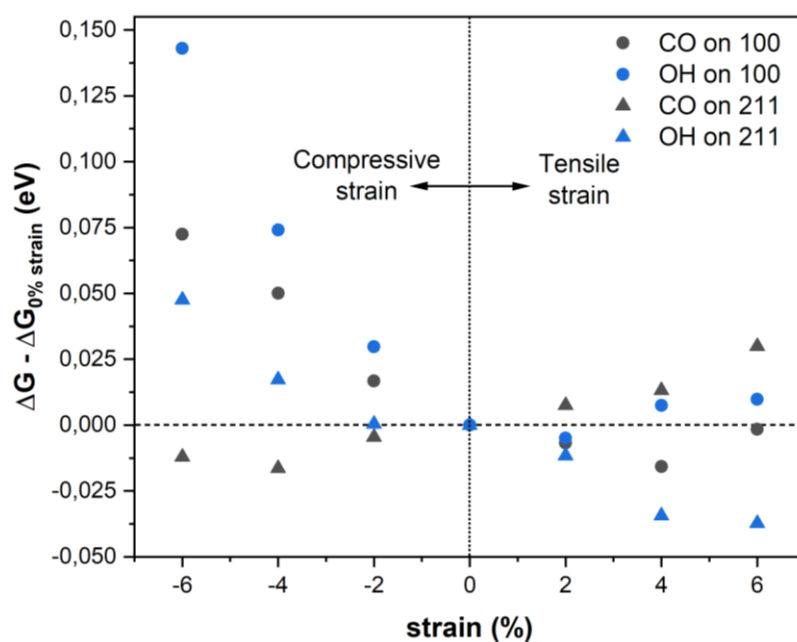


Figure S19 DFT calculated binding energies of $^*\text{CO}$ and $^*\text{OH}$ on the (100) and (211) facets of Cu at varying amounts of strain on the crystal.

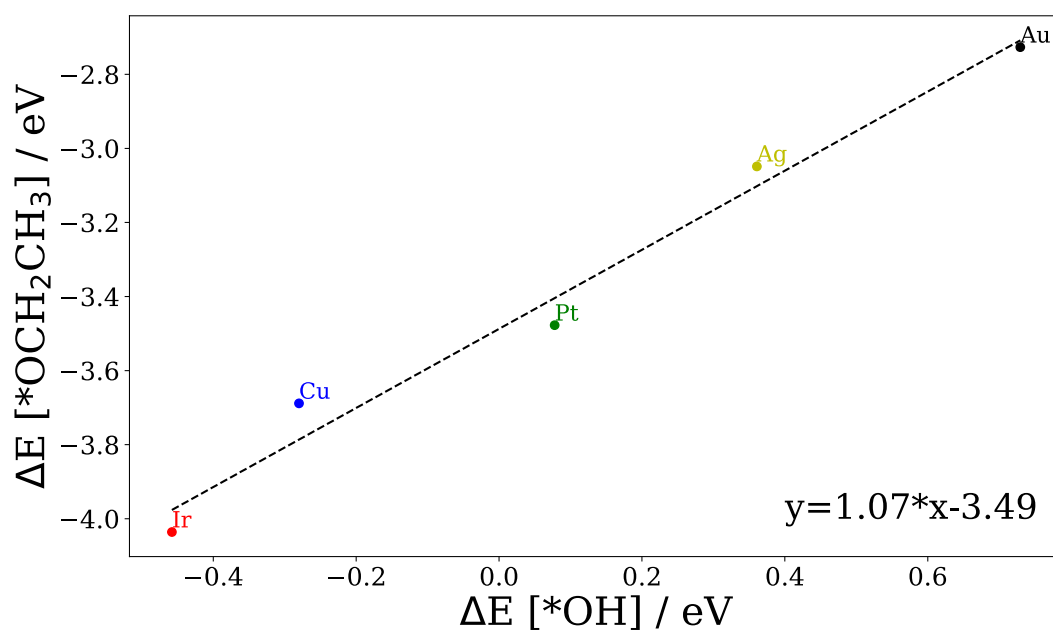


Figure S20 DFT calculated scaling line of $^*\text{OCH}_2\text{CH}_3$ with $^*\text{OH}$, justifying the use of the latter as a descriptor for the binding of oxygenate intermediates.

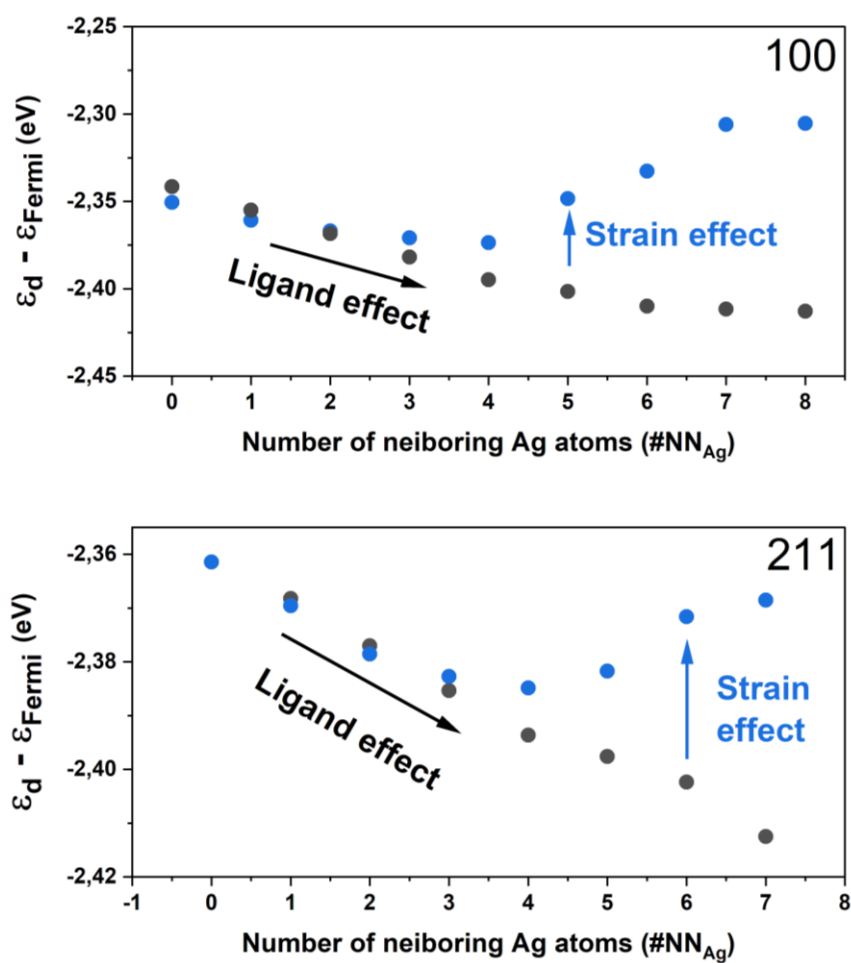


Figure S21 DFT calculated Cu d-band centers in dependency of the number of Ag nearest neighbors.

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